07.09.2023

Mavzu: Modulyar arifmetika. Taqqoslama va uning xossalari. EKUB topishning Yevklid usuli.

Asimmetrik shifrlash algoritmlarining tezligi simmetrik shifrlash algoritmlaridan 1000 barobargacha sekinroq, hajm boʻylcha 30%-100% oraligida hajmni kopaytiradi, simmetrik algoritmlar ma'lumotni shifrlaganda hajmni ozgartirmaydi.

Shuning uchun asimmetrik shifrlash algoritmlari kriptografiyada faqat simmetrik shifrlash algoritmlarining seans kalitlarini generatsiya qilishda ishlatiladi.

DES algoritmi 1974-yildan boshlab tadqiqot ishlari olib borilgan, 1976-yilda ishlab chiqilgan.

AES 2002-yilda AQShda standart algoritm sifatida qabul qilingan.

Biz hozirgacha maxfiy ma'lumotlarni shifrlash uchun ГОСТ 28147-89 (Rossiya) standartidan foydalanamiz. UzDSt dan maxfiy ma'lumotlarni shifrlashga ruxsat berilmaydi.

Biror toʻplamning ixtiyoriy 2 ta elementini olib, ular ustida arifmetik amallar (+, -, *, /) bajarilganda, ~~yana~~ natija yana osha toplamga kelib tushsa, shu holda binar amal bajarilgan boladi.
Masalan, ixtiyoriy 2 ta natural sonni qoshganda yana natural son boladi, ya'ni

natija yana shu toplamga qaytyapti, 2 ta natural sonni biridan ikkinchisini ayirganda natural son bolmasligi m-n.

Taqqoslama xossalari

1. Bir xil modulli taqqoslamalarni hadlab qoshish va ayirish mumkin:

$$a_1 \equiv b_1 \pmod m$$
$$a_2 \equiv b_2 \pmod m$$
$$\vdots$$
$$a_k \equiv b_k \pmod m$$

$$\left[a_1+a_2+\ldots+a_k \equiv b_1+b_2+\ldots+b_k \pmod m\right]$$

$$\left[a_1-a_2-\ldots-a_k \equiv b_1-b_2-\ldots-b_k \pmod m\right]$$

2. Taqqoslamaning bir qismidagi sonni 2-qismiga teskari ishora bilan otkazish mumkin:

$$+\begin{cases} a+b \equiv c \pmod m \\ -a \equiv -a \pmod m \end{cases}$$

$$\left[b = c-a \pmod m\right]$$

3. Taqqoslamaning ixtiyoriy qismiga modulga karrali sonni qoshish m-n:

$$\begin{cases} a \equiv b \pmod m \\ mk \equiv 0 \pmod m), \quad k \in Z \end{cases}$$

$$\left[a+mk \equiv b \pmod m\right]$$

4. Bir xil modulli taqqoslamalarni hadlab kopaytirish m-n:

$$\left[a_1 \cdot a_2 \cdot \ldots \cdot a_k \equiv b_1 \cdot b_2 \cdot \ldots \cdot b_k \pmod m\right]$$

5. Taqqoslamaning ikkala qismini (modulni ōzgartirmay) bir xil musbat butun darajaga kōtarish m-n:

$$a \equiv b \pmod m$$

$$\left[a^k \equiv b^k \pmod m\right]$$

6. Modulni ōzgartirmagan holda taqqoslamaning ikkala qismini bir xil butun songa kopaytirish m-n:

$$ak \equiv bk \pmod m$$

7. Taqqoslamaning ikkala qismini modul bilan ōzaro tub bolgan kopaytuvchiga qisqartirish mumkin.

$$\left[\begin{array}{l} ad \equiv bd \pmod m \\ \text{EKUB}(d, m) = 1. \text{ bolsa} \\ a \equiv b \pmod m \end{array}\right]$$

8. Agar $a \equiv b \pmod m$ bolsa u holda $\text{EKUB}(a, m) = \text{EKUB}(b, m)$ boladi.

9. Quyidagi tengliklar ōrinli:

$$(a \underset{\times}{\pm} b) \bmod n = (a \pmod n) \underset{\times}{\pm} b \pmod n) \bmod n$$

$$a(b+c) \pmod n = ((ab) \pmod n) + (ac) \pmod n)) \pmod n)$$

10. Agar $a \equiv b \pmod{mn}$ bōlsa, u holda $a \equiv b \pmod m$ va $a \equiv b \pmod n$

$(2023^{2022} + 2022^{2023}) \bmod 100$

Ta'rif: Agar 2 ta a va b butun sonp… bir xil r qoldiq tōgri kelsa, ular m modul bōyicha teng qoldiqli yoki m modu… bōyicha taqqoslanadigan sonlar deb ataladi.

$$\left[a \equiv b \pmod m\right]$$

Misol: $\left(2021^{2022} + 2022^{2023}\right) \bmod 100 = ?$

Vechish: 1-usul:

① $\overline{\dots x1}^{\,n} = \overline{\dots (nx)(\mathrm{mod}\ 10)\,1}$

Isbot: Matematik induksiya.

$n=1$: $\overline{\dots x1}^{\,1} = \overline{\dots x1}$

$n=2$: $\overline{\dots x1}^{\,2} = \overline{\dots \underbrace{(2x)(\mathrm{mod}\ 10)}\,1}$

$$\begin{array}{r} \times\ \overline{\dots x1} \\ \overline{\dots x1} \\ \hline \overline{\dots x1} \\ +\ \overline{\dots x} \\ \hline \overline{\dots (2x)(\mathrm{mod}\ 10)\,1} \end{array}$$

$n=k$ da ham ōrinli bōlsin:

$$\overline{\dots x1}^{\,k} = \overline{\dots \underbrace{(kx)(\mathrm{mod}\ 10)}\,1}$$

$n=k+1$: $\overline{\dots x1}^{\,k+1} = \overline{\dots x1}^{\,k} \cdot \overline{\dots x1}$

$$\Downarrow$$

$$\overline{\dots (kx)(\mathrm{mod}\ 10)\,1}$$

$$\begin{array}{r} \overline{\ldots(kx)(\bmod 10)\,1} \\ \times\ \overline{\ldots x1} \\ \hline \overline{\ldots(kx)\,1} \\ +\ \overline{\ldots x} \\ \hline \overline{\ldots((k+1)x)\,1} \end{array} \quad \Delta$$

$$\boxed{2021^{2022} = \ldots 41}$$

(2) $\overline{\ldots 24}^{2n} = \overline{\ldots 76}$

$\overline{\ldots 24}^{2n+1} = \overline{\ldots 24}$

$2022^{2023} = \underline{2^{2023}} \cdot \underline{1011^{2023}}$

$2^{2023} \pmod{100} = ?$

$(2^{10})^{202} \equiv 24^{202} \pmod{100}$

$2^{2020} \equiv 76 \pmod{100}$

$2^{2020} \cdot 2^3 \equiv 76 \cdot 8 \pmod{100}$

$76 \cdot 8 = 608$

$$\boxed{2^{2023} \equiv 8 \pmod{100}}$$

$1011^{2023} \pmod{100} = ?$

$1011^{2023} = \overline{\ldots 31}$

$$\boxed{1011^{2023} \equiv 31 \pmod{100}}$$

$8 \cdot 31 = 248$

$\boxed{248 \equiv 48 \pmod{100}}$

$\boxed{2022^{2023} = \overline{\ldots 48}}$

MC $\overline{\ldots 41} + \overline{\ldots 48} = \overline{\ldots 89}$

Ост. хл. 23й

2-usul: Eyler teoremasi (Eyler funksiyas kelib chiqadi).

$$2021^{2022} + 2022^{2023} \equiv x \pmod{100}$$

Eyler teoremasi:

$$a^{\varphi(m)} \equiv 1 \pmod{m} \rightarrow \text{EKUB}(a, m) = 1$$

$\text{EKUB}(2021; 100) = 1$

$a = 2021, \quad m = 100$

$$\left[2021^{\varphi(100)} \equiv 1 \pmod{100}\right] \quad (1)$$

$$\varphi(100) = \varphi(2^2 \cdot 5^2) = \varphi(2^2) \cdot \varphi(5^2) = (2^2 - 2) \cdot (5^2 - 5) = 2 \cdot 20 = 4$$

$$(1) \rightarrow \quad 2021^{40} \equiv 1 \pmod{100}$$

$$\left[2021^{40} \equiv 1 \pmod{100}\right] \quad (2)$$

~~2022 : 39 = 51 (33)~~ $2022 : 40 = 50 \ (22)$

$$(2) \rightarrow \quad (2021^{40})^{50} \equiv 1^{50} \pmod{100}$$

$$\left[2021^{2000} \equiv 1 \pmod{100}\right]$$

$$\times \quad 2021^{22} \equiv 41 \pmod{100}$$

$$\left[2021^{2022} \equiv 41 \pmod{100}\right] \quad (1)$$

$$2022^{2023} \equiv x \pmod{100} \qquad \text{EKUB}(2022; 100) = 2 \neq 1$$

$$2^{2023} \cdot 1011^{2023} \equiv x \pmod{100} \rightarrow \text{EKUB}(1011, 100) = 1$$

idan

$1011^{\varphi(100)} \equiv 1 \pmod{100}$

$1011^{40} \equiv 1 \pmod{100}$

$1011^{40\cdot 50} \equiv 1^{50} \pmod{100}$

$\times \begin{cases} 1011^{2000} \equiv 1 \pmod{100} \\ 1011^{23} \equiv 31 \pmod{100} \end{cases}$

$\times \begin{cases} 1011^{2023} \equiv 31 \pmod{100} \\ 2^{2023} \equiv 8 \pmod{100} \end{cases}$

$[2022^{2023} \equiv 48 \pmod{100}]$ (2)

1011^{23}

0

(1) + (2) = 41 + 48 = 89

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3-usul: Mod.

$$2021^{2022} + 2022^{2023} \mod 100$$

$2021 \equiv 21 \pmod{100}$
$2021^5 \equiv 21^5 \pmod{100}$
$2021^5 \equiv 1 \pmod{100}$
$2021^{2020} \equiv 1 \pmod{100}$
$2021^{2022} \equiv 41 \pmod{100}$

$2022^{2023} = 2^{2023} \cdot 1011^{2023}$

$1011 \equiv 11 \pmod{100}$
$1011^{10} \equiv 11^{10} \pmod{100}$
$1011^{10} \equiv 1 \pmod{100}$
$1011^{2020} \equiv 1 \pmod{100}$
$1011^{2023} \equiv 11^3 \pmod{100}$
$1011^{2023} \equiv 31 \pmod{100}$

$2 \equiv 2 \pmod{100}$
$2^{10} \equiv 24 \pmod{100}$
$2^{2020} \equiv 76 \pmod{100}$
$2^{2023} \equiv 76 \cdot 8 \pmod{100}$
$2^{2023} \equiv 8 \pmod{100}$

$(41 + 31 \cdot 8) \mod 100$
$(41 + 48) \mod 100$
$89 \mod 100$

13.09.2023

Mavzu: Yevklid algoritmi
kengaytirilgan Yevklid algoritmi

RSA - assim. shifr. alg. (ERI)

1) p, q - tub sonlar
$n = p \cdot q$

2) $\varphi(n) = (p-1)(q-1)$

3) $\exists e$, son (ochiq kalit)
$EKUB(\varphi(n), e) = 1$

4) $\exists d$ - soni (yopiq kalit)
$e \cdot d \equiv 1 \pmod{\varphi(n)}$

5) $M^e \bmod n \equiv C$; C - shifr

Yevklid algoritmi

$EKUB(a; b) = ?$

1) $a = b \cdot q_1 + r_1$ $(r_1 \neq 0)$
$b = r_1 \cdot q_2 + r_2$ $(r_2 \neq 0)$
$r_1 = r_2 \cdot q_3 + r_3$ $(r_3 \neq 0)$
. . . . .
$(r_k = 0)$

$\boxed{EKUB(a; b) = r_{k-1}}$

Kengaytirilgan Yevklid algoritmi.

Teorema. Aylaylik, a va b natural sonlar bo'lib, $d = EKUB(a, b)$ bo'lsin. U holda shunday α va β butun sonlar topiladiki,

$$\left[a \cdot \alpha + b \cdot \beta = d\right]$$

tenglik o'rinli bo'ladi.

$a = b \cdot q_1 + r_1$

$b = r_1 \cdot q_2 + r_2$

$r_1 = r_2 \cdot q_3 + r_3$

$\dots\dots$

$r_{n-3} = r_{n-2} \cdot q_{n-1} + r_{n-1} \rightarrow EKUB(a;b)$

$r_{n-2} = r_{n-1} \cdot q_n$

$r_1 = a \cdot x_1 + b y_1$

$r_2 = a \cdot x_2 + b \cdot y_2$

$r_3 = a \cdot x_3 + b \cdot y_3$

$\dots$

$EKUB(a;b) \leftarrow r_{n-1} = a \cdot x_{n-1} + b \cdot y_{n-1}$

$r_n = 0$

Biz bu yerda

$$x_1, x_2, \dots, x_{n-1} \text{ va } y_1, y_2, \dots, y_{n-1}$$

sonlarini topishimiz kerak. Mazkur sonlar quyidagi formula yordamida topiladi.

$$\left[x_j = x_{j-2} - q_j \cdot x_{j-1}\right] \text{ va } \left[y_j = y_{j-2} - q_j \cdot y_{j-1}\right]$$

$$\left[j = 1, \dots, n-1\right]$$

Vazifa:

$a = 1234 \quad b = 54$ bo'lsa α, β - ?

qoldiqlar	bo'luvchi	x	y
a	$*$	x_{-1}	y_{-1}
b	$*$	x_0	y_0
r_1	q_1	x_1	y_1
r_2	q_2	x_2	y_2
- - -	- - -	- - -	- - -
r_{n-2}	q_{n-2}	x_{n-2}	y_{n-2}
r_{n-1}	q_{n-1}	x_{n-1}	y_{n-1}

$a=1234$ $b=54$ $\alpha, \beta - ?$

$1234 = 54\cdot 22 + 46$

$54 = 46\cdot 1 + 8$

$46 = 8\cdot 5 + 6$

$8 = 6\cdot 1 + 2$

$6 = 2\cdot 3 + 0$

$46 = 1234\cdot 1 - 54\cdot 22$

$8 = 54\cdot 1 - 46\cdot 1 =$
$= 54\cdot 1 - (1234\cdot 1 - 54\cdot 22) =$
$= -1234\cdot \underline{1} + 54\cdot \underline{23}$

$6 = 46 - 8\cdot 5 = (1234 - 54\cdot 22) -$
$- (-1234 + 54\cdot 23)\cdot 5 =$
$= \underline{6}\cdot 1234 - 54\cdot \underline{137}$

$2 = 8 - 6 = (-1234 + 54\cdot 23) -$
$- (2\cdot 1234 - 54\cdot 45) =$
$= -3\cdot 1234 + 68\cdot 54$

$2 = 8 - 6 = (-1234 + 54\cdot 23) - (6\cdot 1234 - 54\cdot 137) =$
$= \underline{-7}\cdot 1234 + \underline{160}\cdot 54$

J: $\alpha = -7$ $\beta = 160$

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goldiqlar	bo'luvchi	x	y
1234			
54			
46	22	1	-22
8	1	-1	23
6	5	6	-137
2	1	-7	160

Hisoblang.

1) A) $37689^{34} \pmod{674}$

B) $9389345^{23} \pmod{51}$

c) $149^{126} \pmod{27}$

2) A) $18! \pmod{162}$

B) $(1+2+3+\ldots+100) \bmod 13$

c) $(1^2+2^2+\ldots+25^2) \bmod 21$

D) $146^{243} \pmod{21}$

Yechish

1) A) $37689^{34} \pmod{674}$

$619^{34} \pmod{674}$

$619 \equiv -55 \pmod{674}$

$619^2 \equiv 3025 \pmod{674}$

$619^2 \equiv 329 \pmod{674}$

$619^3 \equiv 329 \cdot 619 \pmod{674}$

$619^3 \equiv 329 \cdot (-55) \pmod{674}$

$619^3 \equiv -18095 \pmod{674}$

$619^3 \equiv 103 \pmod{674}$

$619^4 \equiv 103 \cdot 619 \pmod{674}$

$619^4 \equiv 401 \pmod{674}$

$619^8 \equiv 401^2 \ldots$

$619^8 \equiv 579$

$619^8 \equiv -95 \pmod{674}$

$619^{16} \equiv 95^2 \pmod{674}$

$619^{16} \equiv 9025 \pmod{674}$

$619^{16} \equiv 263 \pmod{674}$

$619^{17} \equiv 263 \cdot 619 \pmod{674}$

$619^{17} \equiv 263 \cdot (-55) \pmod{674}$

$619^{17} \equiv -14465 \pmod{674}$

$619^{17} \equiv 363 \pmod{674}$

$619^{34} \equiv 363^2 \pmod{674}$

$619^{34} \equiv (-311)^2 \pmod{674}$

$619^{34} \equiv 96711 \pmod{674}$

$619^{34} \equiv 329 \pmod{674}$

B) $9389345^{23} \pmod{51}$

$41^{23} \pmod{51}$

$41 \equiv 41 \pmod{51}$

$41^2 \equiv 1681 \pmod{51}$

$41^2 \equiv 49 \pmod{51}$

$41^2 \equiv -2 \pmod{51}$

$(41^2)^6 \equiv (-2)^6 \pmod{51}$

$41^{12} \equiv 64 \pmod{51}$

$[41^{12} \equiv 13 \pmod{51}]$

$\cancel{(41^{12})^2 \equiv 169}$

$\cancel{41^{24} \equiv 16 \pmod{51}}$

$41^2 \equiv -2 \pmod{51}$

$(41^2)^5 \equiv -2^5 \pmod{51}$

X

$41^{10} \equiv 19 \pmod{51}$

$41^{11} \equiv 19 \cdot 41 \pmod{51}$

$41^{11} \equiv 19 \cdot (-10) \pmod{51}$

$[41^{11} \equiv 14 \pmod{51}]$

$41^{23} \equiv 13 \cdot 14 \pmod{51}$

$41^{23} \equiv 182 \pmod{51}$

$41^{23} \equiv 29 \pmod{51}$

c) $149^{126} \pmod{27}$

$14^{126} \pmod{27}$

$14 \equiv 14 \pmod{27}$

$14^2 \equiv 7 \pmod{27}$

$14^4 \equiv 22 \pmod{27}$

$14^4 \equiv -5 \pmod{27}$

$14^8 \equiv (-5)^2$

$14^8 \equiv -2$

$14^{40} \equiv (-2)^5 \pmod{27}$

$14^{40} \equiv -5$

$14^{80} \equiv 25 \pmod{27}$

$14^{80} \equiv -2$

$14^{120} \equiv -10 \pmod{27}$

$14^{120} \cdot 14^2 \cdot 14^4 \equiv (-10) \cdot 7 \cdot (-5) \pmod{27}$

$\cancel{14^{126} \equiv 350 \ (27)}$

$\cancel{14^{126} \equiv 26 \pmod{27}}$

$14^{126} \equiv 7 \cdot 4 \pmod{27}$

$14^{126} \equiv 1 \pmod{27}$

2) A) $18! \pmod{162}$

$5! \equiv 120 \pmod{162}$

$6! \equiv -42 \pmod{162}$

$6! \equiv -252 \pmod{162}$

$6! \equiv 72 \pmod{162}$

$7! \equiv 18 \pmod{162}$

$8! \equiv 18 \cdot 8$

$8! \equiv -18 \pmod{162}$

$9! \equiv -18 \cdot 9$

$9! \equiv -162 \pmod{162}$

$9! \equiv 0 \pmod{162} \longrightarrow 18! \equiv 0 \pmod{162}$

X

B) $(1+2+\ldots+100) \bmod 13$

① $\frac{1+100}{2} \cdot 100 = 5050$

$5050 \equiv 6 \pmod{13}$

② $\frac{1+91}{2} \cdot 91 \equiv 0 \pmod{13}$

$92+93+\ldots+100 \equiv x \pmod{13}$

$1+2+\ldots+9 \equiv x \pmod{13}$

$45 \equiv x \pmod{13}$

$45 \equiv 6 \pmod{13}$

c) $(1^2+2^2+\dots+25^2) \bmod 21$

① $1^2+2^2+\dots+n^2 = \frac{n(n+1)(2n+1)}{6}$

$1^2+2^2+\dots+25^2 = \frac{25(25+1)(2\cdot 25+1)}{6} = \frac{25\cdot \overset{13}{\cancel{26}}\cdot \overset{17}{\cancel{51}}}{\cancel{6}} = 13\cdot 17\cdot 25$

$13\cdot 17\cdot 25 \equiv 13\cdot 17\cdot 4 \pmod{21}$

$13\cdot 68 \pmod{21}$

$13\cdot 5 \pmod{21}$

$65 \pmod{21} \equiv 2$

② $1^2+2^2+3^2+4^2 \equiv 9 \pmod{21}$

$1^2+\dots+5^2 \equiv 13 \pmod{21}$

$\dots+6^2 \equiv 7$

$\dots+7^2 \equiv 14$

$\dots+8^2 \equiv 15$

$\dots+9^2 \equiv 12$

$\dots+10^2 \equiv 7$

$\dots+11^2 \equiv 2$

$\dots+12^2 \equiv -1$

$\dots+13^2 \equiv 1$

$\dots+14^2 \equiv 8$

$\dots+15^2 \equiv 1$

$\dots+16^2 \equiv 5$

$\dots+17^2 \equiv 0$

$\dots+18^2 \equiv (-3)^2 \equiv 9$

$\dots+19^2 \equiv 9+(-2)^2 \equiv 13$

$\dots+20^2 \equiv 13+(-1)^2 = 14$

$\dots+21^2 \equiv 14$

$\dots+22^2 \equiv 15$

$\dots+23^2 \equiv 19 \equiv -2$

$\dots+24^2 \equiv 7$

$\dots+25^2 \equiv 2$

d) $146^{243} \pmod{21}$

$146 \equiv -1 \pmod{21}$

$146^{243} \equiv -1 \pmod{21}$

$146^{243} \equiv 20 \pmod{21}$

2. XI. 23г.

20.09.2023

Mavzu: Diofant tenglamasi.

Ta'rif: Aytaylik $a, b, c \in \mathbb{Z}$ bolsin. U holda quyidagi korinishdagi tenglamaga **Diofant tenglamasi** deyiladi.

$$[ax+by=c]$$

Teorema: Diofant tenglamasi butun (x_0, y_0) yechimga ega boladi, faqat va faqat $c : EKUB(a, b)$ bolinma butun bolsa, hamda bu yechim quyidagi korinishda:

$$x_0=(\alpha c)/EKUB(a,b), \quad y_0=(\beta c)/EKUB(a,b)$$

bolib, bu yerda α, β - koeffitsiyentlar $EKUB(a,b)=\alpha \underset{a}{x}+\beta \underset{b}{y}$ tenglama yechimlari bolsa.

Teorema: Agar $a, b \neq 0$ butun sonlar, (x_0, y_0)-juftlik esa $ax+by=c$ tenglama yechimi bolsa, u holda ixtiyoriy yechim quyidagi tengliklar bilan

$$x=x_0+bt/d, \quad y=y_0-at/d$$ aniqlanadi.

bu yerda:

t - ixtiyoriy butun son

$d = EKUB(a, b)$

Vazifa.

$85x + 34y = 51 \qquad (x, y) = ?$

Mavzu: Eyler funksiyasi va uning xossalari.

Ta'rif: Quyidagi 2 ta shartni bajaruvchi funksiyaga Eyler funksiyasi d-di.

1) $\varphi(1) = 1$;

2) $\varphi(a)$ funksiya qiymati, a dan kichik va a bilan o'zaro tub bo'lgan sonlarning umumiy soniga teng.

Ta'rif: Natural sonlar to'plamida aniqlangan f-funksiya uchun $EKUB(m, n) = 1$ bo'lganda $f(m \cdot n) = f(m) \cdot f(n)$ tenglik bajarilsa, u holda $f(m \cdot n)$ funksiya multiplikativ funksiya deyiladi

Teorema 1: Eyler funksiyasi multiplikativ funksiyadir.

Teorema 2: p-tub son bölsa, $\varphi(p)=p-1$.

Teorema 3: $a=p_1^{\alpha_1}\cdot p_2^{\alpha_2}\cdot\ldots\cdot p_n^{\alpha_n}$

$$\varphi(a)=a\cdot\left(1-\frac{1}{p_1}\right)\cdot\left(1-\frac{1}{p_2}\right)\cdot\ldots\cdot\left(1-\frac{1}{p_n}\right)$$

Eyler teoremasi: Agar $EKUB(a,m)=1$ bölsa, u holda $\left[a^{\varphi(m)}\equiv 1 \pmod m\right]$ taqqoslama ōrinli.

Diofant tenglamasiga oid misol:

$272x+102y=68$ butun yechimlari topilsin.

Yechish: Diofant tenglamasi butun yechimga ega bölishi u-n:

$c \vdots EKUB(a,b)$, ya'ni $68:34=2$ bajarildi.

U holda Yevklid algoritmiga köra:

$272=102\cdot 2+68$

$102=68\cdot 1+34$

$68=34\cdot 2+0$, ya'ni $EKUB(272,102)=34$.

Kengaytirilgan Yevklid algoritmiga muvofiq esa:

Nazariya har doim amaliyotdan oldin yuradi.

$272 \cdot \alpha + 102 \cdot \beta = 34$

$\alpha = ?$, $\beta = ?$ qiymatlarni topamiz,

$34 = 102 - 68 \cdot 1$

$68 = 272 - 102 \cdot 2$

U holda

$34 = 102 - 68 \cdot 1 = 102 - 1 \cdot (272 - 102 \cdot 2) = 3 \cdot 102 - 272,$

yoki $34 = 102 \cdot 3 + 272 \cdot (-1)$; demak $\beta = 3$, $\alpha = -1$.

Xususiy yechim esa Diofant tenglamasi uchun:

$$x_0 = \frac{\alpha \cdot c}{\text{EKUB}(a,b)} = -\frac{1 \cdot 68}{34} = -2$$

$$y_0 = \frac{\beta \cdot c}{\text{EKUB}(a,b)} = \frac{3 \cdot 68}{34} = 6$$

Vazifa:

$85x + 34y = 51$ $\quad (x; y) \to ?$

Yechish: kengaytirilgan Yevklid algoritmiga kõra:

$85 = 34 \cdot 2 + 17$ (1)

$34 = 17 \cdot 2 + 0$ (2) $\quad \text{EKUB}(85, 34) = 17.$

$$\frac{c}{\text{EKUB}(a,b)} = \frac{51}{17} = 3 \in Z$$

demak tenglama yechimga ega.

$$85 \cdot \alpha + 34 \cdot \beta = 17$$

(1) $\rightarrow 17 = 85 \cdot 1 - 34 \cdot 2 \rightarrow \alpha = 1;\ \beta = -2$

$$x_0 = \frac{\alpha \cdot c}{\text{НСД}(a, b)} = \frac{1 \cdot 51}{17} = 3$$

$$y_0 = \frac{\beta \cdot c}{\text{НСД}(a, b)} = \frac{-2 \cdot 51}{17} = -6$$

$$\boxed{(x_0; y_0) = (3; -6)}$$

$\cancel{x = x_0 + y t/d = 3 +}$

$\cancel{x = x_0 + \beta t/d = 3 + (-2) t}$

$$\text{Ж} \begin{cases} x = x_0 + b t/d = 3 + 34 \cdot t/17 = \underline{3 + 2t} \\ y = y_0 + a t/d = -6 + 85 t/17 = \underline{-6 + 5t} \end{cases}$$

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27.09.2023

Fermaning kichik teoremasi.

Fermaning kichik teoremasi El-Gamal algoritmi parametrlari qatoriga kiruvchi $0<\omega<p-1$, $\omega^{p-1} \equiv 1 mod(p)$, ω sonini aniqlashda, shuningdek, $a^{p-1} (mod\ p)$ qoldiqni p-tub va a-soni p-soniga bolinmaganda sharti ostida hisoblashda tegishli yengilliklar tug'diradi. Quyida mazkur teoremani keltiramiz.

Fermaning kichik teoremasi. Agar a soni p soniga bolinmasa va p tub son bolsa, u holda

$$a^{p-1} \equiv 1 \ (mod\ p)$$

taqqoslama o'rinli boladi.

~~Misol. $a=2$, $n=341$ $\varphi(341)=$~~

Izoh. Shuni ta'kidlash kerakki, teorema bir tomonga to'g'ri, ya'ni

$$a^{p-1} \equiv 1 \ (mod\ p)$$

taqqoslama bajarilsa, har doim ham

p-tub son bo'lmasligi m-n.

Misol. $a=2$, $n=341$, $\varphi(341)=300$

bo'lsin, u holda

$$2^{340} \equiv 1 \pmod{341}$$

taqqoslama o'rinli, lekin 341 murakkab

son, ya'ni $341 = 11 \cdot 31$.

Biroq $2^{10} \equiv 1 \pmod{341}$

bo'lgani uchun: $2^{340} \equiv 1 \pmod{341}$

Berilgan modul bo'yicha teskari elementni topish

Ta'rif: Agar a va n butun sonlari

o'zaro tub bo'lsa, shunday a' son mavjud

bo'lib, $a \cdot a' \equiv 1 \pmod{n}$ taqqoslama

o'rinli bo'ladi.

Bu yerda a' son; a-ga modul

n bo'yicha teskari (element) son d-di.

va $a^{-1} \pmod{n}$ kabi belgilanadi.

Teorema: Bu son quyidagicha topiladi:

$$\left[a^{-1} \equiv a^{\varphi(n)-1} \pmod{n}\right]$$

Bu yerda $\varphi(n)$ - Eyler funksiyasi.

Faqat $EKUB(a,n)=1$ bo'lganda o'rinli boladi.

Chiziqli $a \cdot x \equiv c \pmod{m}$ taqqoslama tenglama yechimini topish.

Teorema 1: $ax \equiv c \pmod{m}$ tenglama butun x-yechimga ega boladi, faqat va faqat $c : EKUB(a, m)$ bo'linma butun bo'lsa. Barcha butun yechimlar esa quyidagicha aniqlanadi:

$$x = x_0 + (tm) / EKUB(a, m)$$

bu yerda $t \in \mathbb{Z}$, (x_0, y_0) esa,

$$ax + my = c$$

tenglama butun yechimi.

Teorema 2: Agar $c : EKUB(a, m)$ bo'linma butun bo'lsa, u holda $ax \equiv c \pmod{m}$ taqqoslama chekli sondagi turlicha yechimlarga ega bo'lib, bu yechimlar quyidagi kŏrinishda boladi:

$$(x_0 + (tm)/EKUB(a,m)) \pmod m$$

$t = 1, 2, \dots, EKUB(a,m)$

Teorema 3: Agar $EKUB(a,c)=1$ bolsa, u holda $ax \equiv b \pmod c$ taqqoslamà:

$$x \equiv a^{\varphi(c)-1} \cdot b \pmod c$$

yechimga ega boladi. $\varphi(c)$ - Eyler funksiyasi

Vazifa:

1) $6x \equiv 7 \pmod{55}$ $x=?$

2) $35x \equiv 14 \pmod{84}$ $x=?$

2 ta usul bilan topish kerak:
1) Eyler 2) $ax \equiv c \pmod m$.

Yechish.

1) $6x \equiv 7 \pmod{55}$

a) Eyler teoremasi bŏyicha teskari elementni topish.

$$x \equiv 7 \cdot 6^{-1} \pmod{55}, \qquad EKUB(6,55)=1$$

$$6^{-1} \pmod{55} = 6^{\varphi(55)-1} \pmod{55}$$

$$\varphi(55) = \varphi(5)\cdot\varphi(11) \cancel{=40} = 4 \cdot 10 = 40$$

$6^{-1} \pmod{55} = 6^{40-1} \pmod{55} = 6^{39} \pmod{55}$

$6 \equiv 6 \pmod{55}$
$6^3 \equiv 216 \pmod{55}$
$6^3 \equiv -4 \pmod{55}$
$6^6 \equiv 16 \pmod{55}$
$6^{12} \equiv 256$
$6^{12} \equiv 36$
$6^{12} \equiv -19$
$\cancel{6^{24} \equiv 361}$
$\cancel{6^{24} \equiv 26}$
$\cancel{6^{36} \equiv 31 \cdot 36 = 806}$
$\cancel{6^{36} \equiv 36}$
$\cancel{6^{39} \equiv -144 \pmod{55}}$
$\cancel{6^{39} \equiv 21 \pmod{55}}$

$6^{12} \equiv -19$
$6^{24} \equiv 361$
$6^{24} \equiv 31 \pmod{55}$
$6^{36} \equiv 31 \cdot 36 = 1116$
$6^{36} \equiv 16 \pmod{55}$
$6^{39} \equiv -64 \pmod{55}$
$\boxed{6^{39} \equiv 46 \pmod{55}}$

$7 \cdot 6^{39} \pmod{55} \equiv 7 \cdot 46 \pmod{55} \equiv 322 \pmod{55} = 47$

$x = 47$

b) Chiziqli taqqoslama tenglama orqali.

$6x \equiv 7 \pmod{55}$

$a = 6 \quad c = 7 \quad m = 55$

$EKUB(6, 55) = 1 \to c : EKUB(a, m) = 7 : 1 = 7 \in Z$.

Demak tenglama yechimga ega.

$6x + 55y = 7$

$EKUB(6; 55) = 1$

$6x + 55y = 1$

$55 = 6 \cdot 9 + 1 \qquad 1 = 55 \cdot 1 - 6 \cdot 9 \Rightarrow \alpha = -9; \ \beta = 1$

$$x_0 = \frac{\alpha \cdot c}{EKUB(a,m)} = \frac{-9 \cdot 7}{1} = -63$$

$$y_0 = \frac{\beta \cdot c}{EKUB(a,m)} = \frac{1 \cdot 7}{1} = 7$$

$$x = x_0 + \frac{m \cdot t}{EKUB(a,m)} = -63 + \frac{55t}{1} = 55t - 63$$

$$x = 55t - 63$$

2) $35x \equiv 14 \pmod{84}$

a) Eyler bo'yicha teskari element topish.

$x \equiv 14 \cdot 35^{-1} \pmod{84}$

$35^{-1} \pmod{84} = 5^{-1} \pmod{12} \cdot 7$

$EKUB(5;12) = 1$

$5^{-1} \pmod{12} = 5^{\varphi(12)-1} \pmod{12}$

$\varphi(12) = \varphi(2^2 \cdot 3) = 2 \cdot 2 = 4$

$5^3 \pmod{12} \equiv 125 \pmod{12} \equiv 5$

$x = 14 \cdot 5 \cdot 7 = 490$

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b) Chiziqli taqqoslama bo'yicha.

$35x \equiv 14 \pmod{84}$

$a = 35 \quad c = 14 \quad m = 84$

$EKUB(a,m) = EKUB(35,84) = 7$

$c : EKUB(a,m) = 14 : 7 = 2 \in Z.$

Demak tenglama yechimga ega

$35x + 84y = 14$

$EKUB(35, 84) = 7$

$35x + 84y = 7$

$84 = 35 \cdot 2 + 14$

$35 = 14 \cdot 2 + 7 \rightarrow 7 = 35 - 14 \cdot 2 = 35 - (84 - 35 \cdot 2) \cdot 2 =$

$= -84 \cdot 2 + 35 \cdot 5$

$14 = 7 \cdot 2 + 0$

$\cancel{\alpha = +5, \quad \beta = 5}$

$\alpha = 5 \quad \beta = -2$

$\cancel{x = x_0 + \frac{\alpha \cdot t}{EKUB(a, m)} =}$

$$x_0 = \frac{\alpha \cdot c}{EKUB(a, m)} = \frac{+5 \cdot 14}{7} = +10$$

$$x = x_0 + \frac{m \cdot t}{EKUB(a, m)} = +10 + \frac{84 \cdot t}{7} = +10 + 12t$$

Ж: $x = 10 + 12t$

04.10.2023

Qoldiqlar haqida Xitoy teoremasi Taqqoslamani hisoblashning effektiv usuli.

Teorema (Qoldiqlar haqida xitoy teoramasi). Aytaylik $m_1, m_2, \ldots, m_n$ juft-jufti bilan ozaro tub sonlar bolsin. U holda ixtiyoriy butun $a_1, a_2, \ldots, a_n$ sonlar uchun

$$x \equiv a_1 \pmod{m_1}$$
$$x \equiv a_2 \pmod{m_2}$$
$$\ldots$$
$$x \equiv a_n \pmod{m_n}$$

taqqoslamalar sistemasi yechimga ega bolib, bu yechim:

$$x = \left(\sum_{j=1}^{n} M_j z_j\right) \pmod{M},$$

bu yerda: $M_j = \left(\prod_{i=1}^{n} m_i\right) / m_j$;

z_j - esa quyidagi taqqoslama

yechimi: $M_j \cdot z_j \equiv a_j \pmod{m_j}$

har bir j uchun va $M = m_1 m_2 \ldots m_n$;

Misol: $2^{6754} \equiv 1 \pmod{3}$

$2^{6754} \equiv 4 \pmod{5}$

$2^{6754} \equiv 2 \pmod{7}$

$2^{6754} \equiv 5 \pmod{11}$

$$\longleftarrow \begin{cases} x \equiv 1 \pmod{3} \\ x \equiv 4 \pmod{5} \\ x \equiv 2 \pmod{7} \\ x \equiv 5 \pmod{11} \end{cases}$$

$\Downarrow$

$2^{6754} \pmod{1155} = ?$

$M_1 = \left(\prod_{i=1}^{4} m_i\right) / m_1 = m_2 \cdot m_3 \cdot m_4 = 5 \cdot 7 \cdot 11 = 385$

$M_2 = 3 \cdot 7 \cdot 11 = 231$
$M_3 = 3 \cdot 5 \cdot 11 = 165$
$M_4 = 3 \cdot 5 \cdot 7 = 105.$

$$M_j \cdot z_j \equiv a_j \pmod{m_j}$$

$M_1 \cdot z_1 \equiv a_1 \pmod{m_1}$
$385 \cdot z_1 \equiv 1 \pmod{3}$
$z_1 = 1$ $\quad z_1 \equiv 385^{-1} \pmod{3} \equiv 385^{\overset{=2}{\varphi(3)}-1} \pmod{3} = 1$

$M_2 \cdot z_2 \equiv a_2 \pmod{m_2}$
$231 \cdot z_2 \equiv 4 \pmod{5}$
$z_2 = 4$ $\quad z_2 \equiv 4 \cdot 231^{-1} \pmod{5} \equiv 4 \cdot 231^{\varphi(5)-1} \equiv 4 \cdot 231^{3} \pmod{5} = 4$

$M_3 \cdot z_3 \equiv a_3 \pmod{m_3}$
$165 \cdot z_3 \equiv 2 \pmod{7}$
$z_3 \equiv 2 \cdot 165^{-1} \pmod{7} \equiv 2 \cdot 165^{\overset{=6}{\varphi(7)}-1} \pmod{7} \equiv$
$z_3 = 4$ $\quad \equiv 2 \cdot 165^{5} \pmod{7} \equiv 2 \cdot 4^{5} \pmod{7} \equiv 2 \cdot \underset{=2}{16} \cdot \underset{=2}{16} \cdot 4 \pmod{7} = 4$

$M_4 \cdot z_4 \equiv a_4 \pmod{m_4}$
$105 \cdot z_4 \equiv 5 \pmod{11}$
$z_4 \equiv 5 \cdot 105^{-1} \pmod{11} \equiv 5 \cdot 105^{\varphi(11)-1} \pmod{11} \equiv$
$\equiv 5 \cdot 105^{9} \pmod{11} \equiv 5 \cdot 6^{9} \pmod{11} = 5 \cdot 36^{4} \cdot 6 \pmod{11}$

$\equiv 5 \cdot 3^4 \cdot 6 \pmod{11} \equiv 5 \cdot 4 \cdot 6 \pmod{11} = 10$ $\quad z_4 = 10$

$x = \left(\sum_{j=1}^{4} M_j \cdot z_j\right) \pmod{M}$ $\quad M = 1155$

$x = (385 \cdot 1 + 231 \cdot 4 + 165 \cdot 4 + 105 \cdot 10) \pmod{1155}$

$x = (385 + 924 + 660 + 1050) \pmod{1155}$

$x = 3019 \pmod{1155}$

$x = 709$

$$\left[\, 2^{6754} \equiv 709 \pmod{1155} \,\right]$$

Taqqoslama $a^e (mod\ n)$ ifodani hisoblashning effektiv usuli.

1-qadam: Daraja kõrsatkichidagi e-soni ikkilik sanoq sistemasiga õtkaziladi:

$$e = b_m \cdot 2^m + b_{m-1} \cdot 2^{m-1} + \ldots + b_1 2^1 + b_0 = [b_m b_{m-1} \ldots b_1 b_0]_2$$

bu yerda $b_i = 0$ yoki 1 va $b_m = 1$

2-qadam: $p_m \equiv a (mod\ n)$ qiymatni yuklaymiz

3-qadam: Barcha $k = m-1; m-2, \ldots, 2, 1$ va $k=0$ qiymatlar uchun

$$p_k = \begin{cases} p_{k+1}^2 (mod\ n), \text{ agar } b_k = 0 \text{ bõlsa} \\ p_{k+1}^2 a (mod\ n), \text{ agar } b_k = 1 \text{ bõlsa} \end{cases}$$

4-qadam: 3-qadamning oxirgi natijasi $p_0 \equiv a^e (mod\ n)$ qidirilayotgan yechim deb e'lon qilinadi.

Misol: Quyidagi $43^{1032} (mod\ 41)$-qoldiqni hisoblang.

Yechishi: Avvalo $e = 1032$ sonni ikkilik sanoq sistemasiga õtkazamiz:

$$1032 = 1024 + 8 = 2^{10} + 2^3 = 10000001000_2$$

Ya'ni $m = 10$, $(b_{10} b_9 b_8 b_7 b_6 b_5 b_4 b_3 b_2 b_1 b_0)_2 =$

$$= (10000001000)_2$$

Demak algoritm qadamiga muvofiq:

$$P_m,\ m = 10, 9, \ldots, 1, 0.$$

Qiymatlar quyidagi qoida bo'yicha hisoblanadi:

$$P_{10} = 43 \pmod{41} = 2$$
$$P_9 = 2^2 \pmod{41} = 4$$
$$P_8 = 4^2 \pmod{41} = 16$$
$$P_7 = 16^2 \pmod{41} = 10$$
$$P_6 = 10^2 \pmod{41} = 18$$
$$P_5 = 18^2 \pmod{41} = 37$$
$$P_4 = 37^2 \pmod{41} = 16$$
$$P_3 = 16^2 \cdot 43 \pmod{41} = 20$$
$$P_2 = 20^2 \pmod{41} = 31$$
$$P_1 = 31^2 \pmod{41} = 18$$
$$P_0 = 18^2 \pmod{41} = 37$$

Javob: $43^{1032} \pmod{41} = 37$

Vazifa:

1) $3^{275} \pmod{100}$

2) $6^{5000} \pmod{1000}$

3) $11^{24681} \pmod{83}$

Yechish:

1) $3^{275} \pmod{100}$

$$275 = 256 + 16 + 2 + 1 = \overset{8}{1}\overset{7}{0}\overset{6}{0}\overset{5}{0}\overset{4}{1}\overset{3}{0}\overset{2}{0}\overset{1}{1}\overset{0}{1}{}_2$$

$P_8 = 3 \pmod{100} = 3$
$P_7 = 3^2 \pmod{100} = 9$
$P_6 = 9^2 \pmod{100} = 81$
$P_5 = 81^2 \pmod{100} = 61$
$P_4 = 61^2 \cdot 3 \pmod{100} = 63$
$P_3 = 63^2 \pmod{100} = 69$
$P_2 = 69^2 \pmod{100} = 61$
$P_1 = 61^2 \cdot 3 \pmod{100} = 63$
$P_0 = 63^2 \cdot 3 \pmod{100} = 07$

$$3^{275} \pmod{100} = 07.$$

2) $6^{5000} \pmod{1000} = ?$

$$5000 = 4096 + 512 + 256 + 128 + 8 =$$

$$= \overset{12}{1}\,\overset{11}{0}\,\overset{10}{0}\,\overset{9}{1}\,\overset{8}{1}\,\overset{7}{1}\,\overset{6}{0}\,\overset{5}{0}\,\overset{4}{0}\,\overset{3}{1}\,\overset{2}{0}\,\overset{1}{0}\,\overset{0}{0}{}_2$$

$P_{12} = 6 \pmod{1000} = 6$
$P_{11} = 6^2 \pmod{1000} = 36$
$P_{10} = 36^2 \pmod{1000} = 296$
$P_9 = 296^2 \cdot 6 \pmod{1000} = 696$
$P_8 = 696^2 \cdot 6 \pmod{1000} = 496$
$P_7 = 496^2 \cdot 6 \pmod{1000} = 96$
$P_6 = 96^2 \pmod{1000} = 216$
$P_5 = 216^2 \pmod{1000} = 656$
$P_4 = 656^2 \pmod{1000} = 336$
$P_3 = 336^2 \cdot 6 \pmod{1000} = 896 \cdot 6 \pmod{1000} = 376$
$P_2 = 376^2 \pmod{1000} = 376$
$P_1 = 376^2 \pmod{1000} = 376$
$P_0 = 376^2 \pmod{1000} = 376$

$$6^{5000} \pmod{1000} = 376$$

3) $11^{24681} \pmod{83} = ?$

$$24681 = \overset{14}{16384} + \overset{13}{8192} + \overset{6}{64} + \overset{5}{32} + \overset{3}{8} + \overset{0}{1}$$

$$\begin{array}{ccccccccccccccc} 14 & 13 & 12 & 11 & 10 & 9 & 8 & 7 & 6 & 5 & 4 & 3 & 2 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 1_2 \end{array}$$

$P_{14} = 11 \pmod{83} = 11$
$P_{13} = 11^2 \cdot 11 \pmod{83} = 3$
$P_{12} = 3^2 (83) = 9$
$P_{11} = 9^2 \pmod{83} = 81$
$P_{10} = 81^2 \pmod{83} = 4$
$P_9 = 4^2 \pmod{83} = 16$
$P_8 = 16^2 \pmod{83} = 7$
$P_7 = 7^2 \pmod{83} = 49$
$P_6 = 49^2 \cdot 11 \pmod{83} = 77 \cdot 11 \pmod{83} = 17$
$P_5 = 17^2 \cdot 11 \pmod{83} = 40 \cdot 11 \pmod{83} = 25$
$P_4 = 25^2 \pmod{83} = 44$
$P_3 = 44^2 \cdot 11 \pmod{83} = 27 \cdot 11 \pmod{83} = 48$
$P_2 = 48^2 \pmod{83} = 63$
$P_1 = 63^2 \pmod{83} = 68$
$P_0 = 68^2 \cdot 11 \pmod{83} = 59 \cdot 11 \pmod{83} = 68$

$$11^{24681} \pmod{83} = 68$$

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11.10.2023.

5-mavzu:

Kvadrat taqqoslama. Lejandr, Yakobi simvollari va ularning xossalari.

Ta'rif: Ushbu $ax^2+bx+c\equiv 0 \pmod{m}$, (1) ko'rinishidagi taqqoslama kvadratik taqqoslama deyiladi. Bu yerda a soni m ga bo'linmaydi.

Teorema: (1) ko'rinishdagi kvadratik taqqoslamani har doim

$$x^2\equiv d \pmod{m_1}$$

ko'rinishga keltirish mumkin.

Misol: $4x^2-11x-3\equiv 0 \pmod{13}$

Yechishi: Quyidagi $11\equiv 24 \pmod{13}$, $3\equiv 16 \pmod{1}$ taqqoslamadan $4x^2-24x-16\equiv 0 \pmod{13}$ tenglik o'rinli bo'ladi. Taqqoslama xossasiga ko'ra: EKUB$(4;13)=1$ bo'lganidan 4 ga qisqartirishimiz mumkin, ya'ni

$x^2-6x-4\equiv 0 \pmod{13}$ yoki $(x-3)^2-13\equiv 0 \pmod{13}$

oxirgi taqqoslama $(x-3)^2\equiv 0 \pmod{13}$

teng kuchli bo'ladi.

Demak, javob: $x\equiv 3 \pmod{13}$

Teorema (Eyler teoremasi). Agar EKUB$(a,p)=1$ bo'lib, $a^{(p-1)/2}\equiv 1 \pmod p$ o'rinli bo'lsa, taqqoslama 2 ta yechimga ega bo'ladi.

$a^{(p-1)/2}\equiv -1 \pmod p$ o'rinli bo'lganda esa taqqoslama birorta ham yechimi yo'q.

Biroq, ushbu kriteriyaning kamchilik tomoni shundaki, $x^2\equiv a \pmod p$, EKUB$(a,p)=1$ taqqoslama p-tub son yetarlicha katta bo'lsa, kvadratik taqqoslamani yechish masalasi murakkablashadi. Bunday hollarda $L(a/p)$ ko'rinishda belgilanuvchi va Lejandr simvoli deb atalovchi quyidagi simvoldan foydalaniladi.

Ta'rif: Quyidagi shartlarni qanoatlantiruvchi simvol:

$$L(a/p)=\begin{cases} 1, & \text{agar } a \text{ son } p \text{ toq tub modul bo'yicha kvadratik chegirma bo'lsa;} \\ -1, & \text{agar } a \text{ son } p \text{ toq tub modul bo'yicha kvadrat chegirma bo'lmasa.} \end{cases}$$

Lejandr simvoli deb ataladi. Bu yerda a soni Lejandr simvolining surati, p esa uning maxraji d-di.

Lejandr simvolining xossalari:

1. Agar $a \equiv a_1 (mod\ p)$ bo'lsa, $L(a/p)=L(a_1/p)$ tenglik o'rinli bo'ladi.

2. Agar $L(a/p)$ simvolda $a=1$ bo'lsa, $L(1/p)=1$ bo'ladi.

3. $L((ab\ldots t)/p)=L(a/p)*L(b/p)\cdot\ldots\cdot L(t/p)$

yoki $L(ab^2/p) = L(a/p)$ tenglik o'rinli

4. Agar p va q sonlar toq tub sonlar bo'lsa,

$$L(p/q) = (-1)^{(p-1)/2 \cdot (q-1)/2} \cdot L(q/p)$$

5) $L\left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}}$

6) $L\left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}}$

7) $L\left(\frac{2^2}{p}\right) = 1$

Vazifa.

A) $L(401/757)$ B) $L(215/761)$

C) $L(113/997)$ D) $L(514/1093)$

Yechish

A) $L\left(\frac{401}{757}\right) = (-1)^{(757-1)/2 \cdot (401-1)/2} \cdot L\left(\frac{757}{401}\right) =$

$= (-1)^{378 \cdot 200} \cdot L\left(\frac{757}{401}\right) = L\left(\frac{356}{401}\right) = L \cdot \left(\frac{89 \cdot 2^2}{401}\right) = L\left(\frac{89}{401}\right) =$

$= L\left(\frac{401}{89}\right) = L\left(\frac{45}{89}\right) = L\left(\frac{5 \cdot 3^2}{89}\right) = L\left(\frac{5}{89}\right) = L\left(\frac{89}{5}\right) =$

$= L\left(\frac{4}{5}\right) = L\left(\frac{1 \cdot 2^2}{5}\right) = L\left(\frac{1}{5}\right) = 1$

B) $L\left(\frac{215}{761}\right) = L\left(\frac{5\cdot 43}{761}\right) = L\left(\frac{5}{761}\right)\cdot L\left(\frac{43}{761}\right) =$

$= L\left(\frac{1}{5}\right)\cdot L\left(\frac{761}{43}\right) = 1\cdot L\left(\frac{30}{43}\right) = L\left(\frac{2}{43}\right)\cdot L\left(\frac{3}{43}\right)\cdot L\left(\frac{5}{43}\right) =$

$= \underbrace{(-1)\cdot(-1)}_{+}\cdot L\left(\frac{43}{3}\right)\cdot L\left(\frac{43}{5}\right) = L\left(\frac{1}{3}\right)\cdot L\left(\frac{3}{5}\right) = 1\cdot L\left(\frac{2}{3}\right) = -1$

C) $L\left(\frac{113}{997}\right) = L\left(\frac{997}{113}\right) = L\left(\frac{93}{113}\right) = L\left(\frac{3}{113}\right)\cdot L\left(\frac{31}{113}\right) =$

$= L\left(\frac{113}{3}\right)\cdot L\left(\frac{113}{31}\right) = L\left(\frac{2}{3}\right)\cdot L\left(\frac{20}{31}\right) = (-1)\cdot L\left(\frac{5\cdot 2^2}{31}\right) =$

$= (-1)\cdot L\left(\frac{5}{31}\right) = (-1)\cdot L\left(\frac{31}{5}\right) = (-1)\cdot L\left(\frac{1}{5}\right) = -1$

D) $L\left(\frac{514}{1093}\right) = L\cdot\left(\frac{2\cdot 257}{1093}\right) = L\left(\frac{2}{1093}\right)\cdot L\left(\frac{257}{1093}\right) =$

$= (-1)^{\frac{1093^2-1}{8}}\cdot L\left(\frac{1093}{257}\right) = (-1)^{\frac{1092\cdot 1094}{8}}\cdot L\left(\frac{65}{257}\right) =$

$= (-1)\cdot L\left(\frac{5}{257}\right)\cdot L\left(\frac{13}{257}\right) = (-1)\cdot L\left(\frac{257}{5}\right)\cdot L\left(\frac{257}{13}\right) =$

$= (-1)\cdot L\left(\frac{2}{5}\right)\cdot L\left(\frac{10}{13}\right) = (-1)\cdot(-1)\cdot L\left(\frac{2}{13}\right)\cdot L\left(\frac{5}{13}\right) =$

$= (-1)\cdot L\left(\frac{13}{5}\right) = (-1)\cdot L\left(\frac{3}{5}\right) = (-1)\cdot L\left(\frac{2}{3}\right) = 1$

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18.10.2023

Yakobi simvoli

Lejandr simvolini tezroq hisoblash uchun, bu alomatga nisbatan umumiy bo'lgan Yakobi simvolidan foydalaniladi.

Agar $p = p_1 \cdot p_2 \cdot \ldots \cdot p_k$ bo'lsa, $EKUB(a, p) = 1$ bo'lsa u holda (p - murakkab son, 1 dan katta toq son)

$$j\left(\frac{a}{p}\right) = \left(\frac{a}{p_1}\right)\left(\frac{a}{p_2}\right) \cdot \ldots \left(\frac{a}{p_k}\right)$$

tenglik Yakobi simvoli deyiladi.

Xossalari:

1) Agar $a \equiv a_1 \pmod p$ bo'lsa, $j(a/p) = j(a_1/p)$
2) Agar $j(a/p)$ da $a = 1$ bo'lsa, $j(1/p) = 1$ tenglik o'rinli boladi.
3) $j((a \cdot b \cdot \ldots t)/p) = j(a/p) \cdot j(b/p) \cdot \ldots j(t/p)$ yoki $j(ab^2/p) = j(a/p)$ tenglik o'rinli.
4) Agar P va Q - musbat toq va o'zaro tub sonlar bo'lsa,

$$j(Q/P) = (-1)^{(P-1)/2 \cdot (Q-1)/2} \cdot j(P/Q)$$ tenglik bajariladi.

5) $j\left(\frac{-1}{p}\right)=(-1)^{\frac{p-1}{2}}$

6) $j\left(\frac{2}{p}\right)=(-1)^{\frac{p^2-1}{8}}$

Izoh: Agar Yakobi simvolining maxraji tub son bo'lsa u Lejandr simvoli bilan bir xil bo'ladi va taqqoslama yechimga ega bo'lsa, Yakobi simvoli 1 ga, yechimga ega bo'lmasa −1 ga teng bo'ladi. Lekin Yakobi simvoli tarkibiy modul uchun taqqoslama yechimga ega bo'lmasa ham uning qiymati 1 ga teng bo'lishi mumkin.

Kvadratik taqqoslama yechimlari.

1) $x^2 \equiv a \pmod{p}$, p − tub son (1)

2) $x^2 \equiv a \pmod{m}$, m − murakkab son, (2)

Teorema 1. Agar (1) taqqoslamada $p \equiv 3 \pmod 4$ → $p=4\cdot m+3$ bo'lsa va Lejandr simvoli $L\left(\frac{a}{p}\right)=1$ bo'lsa, u holda kvadratik taqqoslamaning yechimlari

$x \equiv \pm a^{m+1} \pmod p$ (3) bo'ladi.

Teorema 2. Agar $p \equiv 5 \pmod 8$ ya'ni $p = 8 \cdot m + 5$, va $L\left(\frac{a}{p}\right) = 1$ bo'lsa, u holda (1) taqqoslama tenglama yechimi:

$x \equiv \pm a^{m+1} \cdot 2^{m+1} \pmod p$ bo'ladi.

$a^{2m+1} \equiv 1 \pmod p$
$\Downarrow$
$x = \pm a^{m+1} (p)$

$a^{2m+1} \equiv -1 \pmod p$
$\Downarrow$
$x = \pm a^{m+1} \cdot 2^{2m+1} \pmod p$

Misol 1. $x^2 \equiv 7 \pmod{31}$

Misol 2. $x^2 \equiv 10 \pmod{53}$

Misol 1.

$31 = 4 \cdot 7 + 3 \rightarrow 4m+3$

$L\left(\frac{7}{31}\right) = -L\left(\frac{31}{7}\right) = -L\left(\frac{3}{7}\right) = -L\left(\frac{3}{7}\right) = \cancel{1}$

$= +L\left(\frac{7}{3}\right) = L\left(\frac{1}{3}\right) = 1$.

$x \equiv \pm 7^{8} \pmod{31}$

$7^{8} \pmod{31} = ?$

$a = 7 \quad m = 31 \rightarrow$ tub son

$7^{31-1} \equiv 1 \pmod{31}$

$7^{30} \equiv 1 \pmod{31}$

$7^{32} \equiv 18 \pmod{31}$

$7^2 \equiv 18 \pmod{31}$

$7^4 \equiv 14 \pmod{31}$

$7^8 \equiv 10 \pmod{31}$

~~$x = \pm(31k + 10)$~~

$x = \pm(31k + 10) \quad k \in Z$.

Umumiy algoritm:

$x^2 \equiv a \pmod{p}$ p-tub son $(p>2)$ (1)

~~1-qadam~~ shunday $\exists a$ va N sonlari tanlanadiki.

$L\left(\frac{a}{p}\right)=1$ va $L\left(\frac{N}{p}\right)=-1$.

1-qadam: $p=2^k \cdot h+1$ h-toq son

2-qadam: $a_1 = a^{\frac{h+1}{2}} \pmod{p}$, $a_2 = a^{-1} \pmod{p}$,

$N_1 = N^h \pmod{p}$, $N_2 = 1$, $j=0$.

3-qadam: $i=0, 1, 2, \dots, k-2$ uchun

3.1. $b = a_1 \cdot N_2 \pmod{p}$

3.2. $c = a_2 \cdot b^2 \pmod{p}$

3.3. $d = c^{2^{k-2-i}} \pmod{p}$, bu yerda

$d=1$ bo'lganda $j_i=0$ yuklansin.

$d=-1$ bo'lganda $j_i=1$ yuklansin.

3.4. $N_2 = N_2 \cdot N_1^{2^i \cdot j_i} \pmod{p}$

4-qadam:

$x \equiv \pm a_1 \cdot N_2 \pmod{p}$

Misol: $x^2 \equiv 14 \pmod{193}$ $x=?$

$N=5$, $h=3$, $L\left(\frac{N}{p}\right)=-1$ $(193=2^6 \cdot 3+1)$

Teorema 3.

1) $x^2 \equiv a \pmod{m}$, m - murakkab son, (2)

$m = m_1 \cdot m_2 \cdot \ldots \cdot m_n$, EKUB$(m_i, m_j) = 1$ bo'lsin.

U holda (2) taqqoslama tenglama yechimga ega bo'lishi uchun:

$x^2 \equiv a \pmod{m_1}$,

$x^2 \equiv a \pmod{m_2}$,

....

$x^2 \equiv a \pmod{m_n}$ - tenglamalar (taqqoslamalar) yechimga ega bo'lishi zarur va yetarli.

Misol. $x^2 \equiv 13 \pmod{35}$, $\exists x = ?!$ (yechim yo'q)

lekin $J\left(\frac{13}{35}\right) = 1$

i	$b \equiv a_1 N_2$	$c \equiv a_2 b^2$	$d \equiv c^{2^{k-2}}$	j_i	N_2
0	3	42	-1	1	125
1	182	50	-1	1	158
2	88	112	1	0	158
3	88	112	-1	1	39
4	117	192	-1	1	122

Vazifa.

Yechish. $x^2 \equiv 14 \pmod{193}$ $\quad x = ?$

$N = 5$

$L\left(\frac{14}{193}\right) = 1 \qquad L\left(\frac{5}{193}\right) = -1.$

1-q: $p = 2^k \cdot h + 1$

$193 = 2^k \cdot h + 1 \rightarrow k = 6, \; h = 3$

2-q: $a_1 = a^{\frac{h+1}{2}} \pmod{p}$ $\qquad N_1 = N^h \pmod{p} = 5^3 \pmod{193}$

$N_1 = 125$

$a_1 \equiv 14^{\frac{3+1}{2}} \equiv 14^2 \pmod{193} = 3. \qquad a_1 = 3$

$a_2 = a^{-1} \pmod{p}$

$a_2 = 14^{-1} \pmod{193} = 69 \qquad (14^{191} \pmod{193} = 69)$

3-q: $i = 0, 1, 2, 3, 4. \qquad N_2 = 1, \; j = 0$

3.1) $b = a_1 \cdot N_2 \pmod{p}$

$b = 3 \cdot 1 \pmod{193} = 3$

3.2) $c = a_2 \cdot b^2 \pmod{p} = 69 \cdot 9 \pmod{193} = 42$

3.3) $d = 42^{2^4} = 42^{16} \pmod{193} = -1$

$j_0 = +1$

3.4) $N_2 = N_2 \cdot N_1^{2^i \cdot j_i} \pmod{193} = 1 \cdot 125^{2^0 \cdot 1} \pmod{193} =$
$= 125.$

3.1) $b = 182$
3.2) $c = 50$
3.3) $d = -1, \; j_1 = 1$
3.4) $N_2 = 158$

3.1) $b = 88$
3.2) $c = 112$
3.3) $d = 1$, $j_2 = 0$
3.4) $N_2 = 158$

3.1) $b = 88$
3.2) $c = 112$
3.3) $d = -1$, $j_3 = 1$
3.4) $N_2 = 39$

3.1) $b = 117$
3.2) $c = 192$
3.3) $d = -1$, $j_4 = 1$
3.4) $N_2 = 122$

$$x = \pm a_1 \cdot N_2 \pmod p$$

$$x = \pm 3 \cdot 122 \pmod{193} = \pm 173$$

Ж: $x = \pm 173$

2. XI. 23г.

25.10.2023

Matematik strukturalar. Gruppa, xalqa, maydon va uning xossalari.

Kriptografik algoritmlar va protokollar ma'lumotni son yoki chekli fazo elementlari sifatida qayta ishlaydi. Mazkur holda shifrlash va ochiq matnga ega bo'lish ~~jaro~~ (deshifrlash) jarayonida foydalaniladigan amallar berilgan ma'lumotni bir turdan boshqa turga o'tkazadi hamda chekli fazoda yopiqlik (binar amal) xossasiga ega bo'lishi lozim. Biroq sonlar ustidagi oddiy arifmetik amallar (+, -, *, /) ushbu xossani bajarmaydi. Shuning uchun kriptografik algoritmlarda chekli fazodagi ma'lumotlar amalda sonlar ustidagi oddiy arifmetik amallarga keltirilmaydi. Qoidaga asosan ularda algebraik struktura aniqlangan va yopiqlik xossasini bajaruvchi bazolarda amalga oshradi.

Ta'rif: Ixtiyoriy A toplamning tayin bir tartibda olingan $\forall x, y$ - elementiga shu toplamning aniq bir elementi mos qoyilsa, A toplamda binar algebraik amal aniqlangan deyiladi.

Binar amal:

$\cdot, \circ, \top, \oplus, \dots$ ($*$ - binar toplam)

kabi belgilanadi.

$a \circledR b = ?$ $\to$ a ni b ga R parametrli qoshish

Ta'rif: Aytaylik A toplamda biror $*$ binar amal aniqlangan bolsin. Agar $\forall x, y, z \in A$ uchun $(x * y) * z = x * (y * z)$ bolsa, $*$ binar amal assosiativ d-di.

Ta'rif: Agar $\forall x, y \in A$ uchun $x * y = y * x$ bolsa, $*$ binar amal kommutativ deyiladi.

Ta'rif: Agar quyidagi shartlar bajarilsa:

1. G to'plamda binar amal $*$ aniqlangan.
2. Binar amal $*$, assosiativ:

$$\forall x, y, z \in G \text{ uchun } (x*y)*z = x*(y*z)$$

3. G-da neytral element "e" mavjud:

$\forall a \in G$ uchun $a*e = e*a = a$ o'rinli bo'lsa.

4. $\forall a \in G$ uchun G-da a-ga simmetrik (teskari) element mavjud:

$$a*a^{-1} = a^{-1}*a = e$$

bu G to'plam gruppa deyiladi.

Ta'rif: Agar gruppa ta'rifida 1- va 2-shartlar bajarilsa G to'plamga yarim gruppa deyiladi.

Ta'rif: Agar gruppa ta'rifida, 1-, 2- va 3-shartlar bajarilsa, u holda bu to'plamga monoid deyiladi.

Gruppa terminini fanga 1-bo'lib, fransuz matematigi Galua 1832-yilda kiritgan.

Misol.

Butun sonlar toplami qoshish amaliga nisbatan gruppa tashkil etadi.

Ta'rif: K toplamda qoshish va (~~ayirish~~) kopaytirish amallari aniqlangan bolib, quyidagi shartlar bajarilgan bolsa:

1) K-toplam qoshish amaliga nisbatan Abel gruppasi tashkil qilsa,

2) Kopaytirish amali assosiativ.

3) Kopaytirish amali qoshish amaliga nisbatan distributiv, ya'ni

$\forall a, b, c \in K$ uchun $a \cdot (b+c) = ab + ac$

U holda $(K, +, \times)$ struktura xalqa deb ataladi.

Misol 2.

Butun sonlar toplamida + va x

xalqa tashkil etadi.

Misol 3

$(n\times n)$ A-matritsa toplam berilgan.

"." -kopaytirish amali. (binar)

qachon gruppa tashkil etadi.

Vazifa.

Misol 1. Yechish: $a, b, c \in Z$ (+) → binar amal

Gruppa:

1) Binar (+) amal aniqlangan.

2) $(a+b)+c = a+(b+c)$ → assosiativ.

3) $e=0,\ e\in Z$

$a+0=0+a=a$

4) $a+(-a)=(-a)+a=0$

Misol 2. Yechish

(+) amali, (*) amali

1) Butun sonlar toplamida aniqlangan (+, *)

2) Assosiativ.

$(a+b)+c=a+(b+c)$, $(a*b)*c = a*(b*c)$

3) (+) uchun $e=0$ $a+0=0+a=a$

(*) uchun $e=1$ $\quad a*1=1*a=a$.

Demak (*) va (+) amallari butun sonlar toplamida xalqa tashkil etadi.

Misol 3.

$A=\begin{pmatrix}1&2\\3&4\end{pmatrix}$ $\quad B=\begin{pmatrix}5&6\\7&8\end{pmatrix}$ $\quad C=\begin{pmatrix}1&3\\2&4\end{pmatrix}$ $\quad$ (*)-binar amal.

1) (*) amali aniqlangan.

2) Assosiativlik $\quad A\cdot(B\cdot C)=(A\cdot B)\cdot C$.

$$A\cdot B=\begin{pmatrix}1&2\\3&4\end{pmatrix}\cdot\begin{pmatrix}5&6\\7&8\end{pmatrix}=\begin{pmatrix}5+14&6+16\\15+28&18+32\end{pmatrix}=\begin{pmatrix}19&22\\43&50\end{pmatrix}$$

$$B\cdot C=\begin{pmatrix}5&6\\7&8\end{pmatrix}\cdot\begin{pmatrix}1&3\\2&4\end{pmatrix}=\begin{pmatrix}5+12&15+24\\7+16&21+32\end{pmatrix}=\begin{pmatrix}17&39\\23&53\end{pmatrix}$$

$$A\cdot(B\cdot C)=\begin{pmatrix}1&2\\3&4\end{pmatrix}\cdot\begin{pmatrix}17&39\\23&53\end{pmatrix}=\begin{pmatrix}17+46&39+106\\51+92&117+212\end{pmatrix}=\begin{pmatrix}63&145\\143&329\end{pmatrix}$$

$$(A\cdot B)\cdot C=\begin{pmatrix}19&22\\43&50\end{pmatrix}\cdot\begin{pmatrix}1&3\\2&4\end{pmatrix}=\begin{pmatrix}19+44&57+88\\43+100&129+200\end{pmatrix}=\begin{pmatrix}63&145\\143&329\end{pmatrix}$$

3) $e=\begin{pmatrix}1&0\\0&1\end{pmatrix}$ $\quad A\cdot e=e\cdot A=A$

4) $A\cdot A^{-1}=A^{-1}\cdot A=e$ (Agar $\det A\neq 0$ bo'lsa)

2.XI.23y.

Misol 2. (+) va (*) xalqa

1) Z toplam (+) amaliga nisbatan Abel gruppa tashkil etadi.

2) (*)-assosiativ $\quad (a*b)*c=a*(b*c)$

3) distributiv $\quad a(b+c)=a\cdot b+a\cdot c$

4) $(Z,+,\times)$ struktura xalqa tashkil etadi.

30.10.2023.

AES algoritmning matematik asoslari

Xalqaning quyidagi xossalari mavjud.

1. Har bir xalqada uning noli deb ataluvchi, yagona θ-element mavjud.

2. Har bir xalqa ixtiyoriy elementi uchun unga qarama-qarshi bo'lgan yagona element mavjud, ya'ni $\forall a \in K$ uchun shunday "$-a$" element topiladiki $a+(-a)=\theta$

3. xalqada (K) $a+x=b$ tenglama yechimga ega va u yagonadir. Bu yechim

$$x=-a+b$$

ga teng

Ta'rif: Agar K-xalqaning ixtiyoriy a va b elementlari uchun $ab = ba$ tenglik bajarilsa, u holda K-xalqa **kommutativ xalqa** deyiladi.

Ta'rif: P-kommutativ xalqa bo'lsin. Agar $\forall a, b \in P$ va $a \neq 0$ uchun $ax = b$ tenglikni qanoatlantiruvchi faqat bitta element $x \in P$ mavjud bo'lsa, u holda P-to'plam maydon deyiladi.

Maydon xossalari.

1. Har bir P maydonda birlik element mavjud va u yagona.
2. $x \neq 0 \in P$ bo'lsa, P-da x-ga teskari yagona x^{-1} element mavjud.
3. Ixtiyoriy P maydon nolining bo'luvchilariga ~~nisbatan~~ ega emas. Ya'ni $\forall x, y \in P$ va $xy = \theta$ bo'lsa, u holda $x = \theta$ yoki $y = \theta$.

Vazifa

Misol. $f(x)=\frac{a_i x+b_i}{c_i x+d_i}$ - kasr chiziqli funksiyalar toplami.

M_f - b-n belgilangan binar amal, quyidagi: $f_1 f_2 = f_1(f_2(x))$ bolsa, u holda toplam gruppami?

Misol 2. Kompleks sonlar toplami odatdagi qoshish va kopaytirish amaliga nisbatan maydon tashkil etishi korsatilsin.

- Kvadrat matritsalar kommutativlik xossasiga ega emas, lekin assosiativlik xossasiga ega.

Kom: $a*b \neq b*a$
Assos: $a*(b*c)=(a*b)*c$.

Mavzu: Keltirilmaydigan kophadlar maydoni.

$$\text{SubBytes}() \Leftrightarrow \begin{cases} S_{ij}\cdot S_{ij}^{-1} \equiv 1 \pmod{\varphi(x)}, \quad \varphi(x)\in GF(2^8) \\ S_{ij}^{-1}=b_i \quad c\, b \oplus b_1 = \_\ \text{bayt} \end{cases}$$

keltirilmaydigan 8-darajali kophad.

Ta'rif: $g(x) \in F(x)$ - ko'phad $f(x) \in F(x)$ ko'phad-ni bo'ladi deyiladi, agar $\exists h(x) \in F(x)$ - ko'phad mavjud bo'lib, $f(x) = g(x) \times h(x)$ o'rinli bo'lsa. Bu yerda $F(x)$ - qandaydir F - maydonda ko'phadlar to'plami deb qaraladi.

Misol 1. $f(x) = 2x^5 + x^4 + 4x + 3$ $\quad g(x) = 3x^2 + 1 \ (\in F_5[x])$

$q(x) = ?$ $\quad r(x) = ? \ (\in F_5[x] = ?)$

$$
\begin{array}{l|l}
2x^5 + x^4 + 4x + 3 & 3x^2 + 1 \\
\hline
-\ 2x^5 + 4x^3 & 4x^3 + 2x^2 + 2x + 1 \\
\quad x^4 - 4x^3 + 4x + 3 & \\
= x^4 + x^3 + 4x + 3 & \\
-\ x^4 + 2x^2 & \\
\quad x^3 - 2x^2 + 4x + 3 & \\
= x^3 + 3x^2 + 4x + 3 & \\
-\ x^3 + 2x & \\
\quad -\ 3x^2 + 2x + 3 & \\
\quad\ \ 3x^2 + 1 & \\
\quad\quad 2x + 2 &
\end{array}
$$

$r(x) = 2x + 2$

$q(x) = 4x^3 + 2x^2 + 2x + 1.$

Vazifa

Misol 3

Ko'rsatilgan maydonda Yevklid usuli bilan EKUB(f, g) topilsin.

A) $f = x^7 + 1$, $g(x) = x^5 + x^3 + x + 1$, $F = F_2$

B) $f = x^5 + x + 1$, $g(x) = x^6 + x^5 + x^4 + 1$, $F = F_2$

c) $f = x^8 + 2x^5 + x^3 + x^2 + 1$, $g(x) = 2x^6 + x^5 + 2x^3 + 2x^2 + 2$, $F = F_3$

Vazifa
Yechish.

Misol 2. kompleks sonlar (+), (*) - binar amallar Maydon tashkil etishi.

1) kompleks sonlar maydonida $i\sqrt{-1}$ birlik element h-di.

2) (*) $x = 2i + 3 \neq 0$ $\qquad x = a+bi$

$x^{-1} = \frac{1}{2i+3}$ $\qquad x^{-1} = \frac{1}{a+bi}$

(+) $\Rightarrow x^{-1} = -x$

3) kompleks sonlar maydonining noli $b=0 \rightarrow x = a+bi = 0.$ $\quad \theta = 0$
$a=0 \nearrow$

$x \cdot y = 0$ bo'lishi uchun x yoki y nolga teng bo'lishi kerak.

Demak Kompleks sonlar maydoni Maydon tashkil etadi.

Misol 3.

A) $f(x) = x^7+1, \quad g(x) = x^5+x^3+x+1 \qquad F = F_2$

$$\begin{array}{l|l} x^7+1 & x^5+x^3+x+1 \\ -\underline{x^7+x^5+x^3+x^2} & x^2+1 \\ -x^5-x^3-x^2+1 & \end{array}$$

mod 2 $\rightarrow x^5+x^3+x^2+1$
$-\underline{x^5+x^3+x^2+1}$
x^2-x

mod 2 $\rightarrow$ x^2+x

$$\begin{array}{l|l} x^5+x^3+x+1 & x^2+x \\ -\underline{x^5+x^4} & x^3+x^2 \\ -x^4+x^3+x+1 & \end{array}$$

mod 2 $\rightarrow x^4+x^3+x+1$
$-\underline{x^4+x^3}$
$x+1 \rightarrow$ EKUB $(f(x), g(x))$

$$\begin{array}{l|l} x^2+x & x+1 \\ -\underline{x^2+x} & x \\ 0 & \end{array}$$

B) $f(x)=x^5+x+1, \quad g(x)=x^6+x^5+x^4+1, \quad F=F_2$

$$\begin{array}{l|l} x^6+x^5+x^4+1 & x^5+x+1 \\ \hline -\,x^6+x^2+x & x+1 \\ \hline x^5-x^4-x^2-x+1 & \\ 2\to x^5+x^4+x^2+x+1 & \\ -\,x^5+x+1 & \\ \hline x^4+x^2 & \end{array}$$

$$\begin{array}{l|l} x^5+x+1 & x^4+x^2 \\ \hline -\,x^5+x^3 & x \\ \hline x^3+x+1 & \end{array}$$

$$\begin{array}{l|l} x^4+x^2 & x^3+x+1 \\ \hline -\,x^4+x^2+x & x \\ \hline x & \end{array}$$

$$\begin{array}{l|l} x^3+x+1 & x \\ \hline -\,x^3 & x^2+1 \\ \hline x+1 & \\ -\,x & \\ \hline 1 & \end{array}$$

$f(x)$ va $g(x)$ ōzaro tub.

c) $f(x)=x^8+2x^5+x^3+x^2+1 \qquad g(x)=2x^6+x^5+2x^3+2x^2+2 \quad F=F_3$

$$\begin{array}{l|l} x^8+2x^5+x^3+x^2+1 & 2x^6+x^5+2x^3+2x^2+2 \\ \hline -\,4x^8+2x^7+4x^5+4x^4+4x^2 & 2x^2+2x+2 \\ \hline -3x^8-2x^7-2x^5-4x^4+x^3-3x^2+1 & \\ 3\to x^7+x^5+2x^4+x^3+1 & \\ -\,4x^7+2x^6+4x^4+4x^3+4x & \\ \hline -3x^7-2x^6+x^5-2x^4-3x^3-4x+1 & \\ 3\to x^6+x^5+x^4+2x+1 & \\ 4x^6+2x^5+4x^3+4x^2+4 & \\ \hline -3x^6-x^5+x^4-4x^3-4x^2+2x-3 & \\ \text{d}\,3\to 2x^5+x^4+2x^3+2x^2+2x & \end{array}$$

$$\begin{array}{l|l} 2x^6+x^5+2x^3+2x^2+2 & 2x^5+x^4+2x^3+2x^2+2x \\ \hline -\,2x^6+x^5+2x^4+2x^3+2x^2 & x \\ \hline -2x^4+2 & \\ \text{mod}\,3\to x^4+2 & \end{array}$$

Kim
2.XI.23й.

$$\begin{array}{l|l} 2x^5+x^4+2x^3+2x^2+2x & x^4+2 \\ -\ 2x^5+4x & \hline 2x+1 \\ \hline x^4+2x^3+2x^2-2x & \\ -\ x^4+2 & \\ \hline 2x^3+2x^2+x+1 & \end{array}$$

$$\begin{array}{l|l} x^4+2 & 2x^3+2x^2+x+1 \\ +\ 4x^4+4x^3+2x^2+2x & \hline 2x+1 \\ \hline -3x^4-4x^3-2x^2-2x+2 & \end{array}$$

mod 3 $\rightarrow$ $2x^2+x$

$$\begin{array}{l|l} 2x^3+2x^2+x+1 & 2x^2+x \\ -\ 2x^3+x^2 & \hline x+2 \\ \hline x^2+x+1 & \\ -\ 4x^2+2x & \\ \hline -3x^2-x+1 & \end{array}$$

mod 3 $\rightarrow$ $2x+1$ $\rightarrow$ EKUB$(f(x), g(x))$

$$\begin{array}{l|l} 2x^2+x & 2x+1 \\ -\ 2x^2+x & \hline x \\ \hline 0 & \end{array}$$

mod 3 $\quad 2x^3+x^2+x+2$

$$\begin{array}{l} 2x^3+2x^2+x+1 \\ \hline -x^2+1 \end{array}$$

mod 3 $\rightarrow$ $2x^2+1$

$$\begin{array}{l|l} 2x^3+2x^2+x+1 & 2x^2+1 \\ -\ 2x^3+x & \hline x+1 \\ \hline 2x^2+1 & \\ -\ 2x^2+1 & \\ \hline 0 & \end{array}$$

EKUB$(f(x), g(x)) = 2x^2+1$

Ta'rif: $f(x) \in F[x]$ ko'phad keltiriladigan ko'phad deb ataladi. F-maydon ustida yoki $F[x]$-xalqada, agar $f(x) = g \cdot h$; $g, h \in F(x)$ tenglik bajariladigan) faqat $g = const$ yoki $h = const$ bo'lganda, ya'ni o'zgarmas ko'phadlar deb tushuniladi.

Misol. F_2 maydondagi 4-darajali barcha keltirilmaydigan ko'phadlarni toping.

$f_1 = x^4 + x + 1$ $\quad f_2(x) = x^4 + x^3 + 1$

$f_3(x) = x^4 + x^3 + x^2 + x$

shu 3 ta ko'phad keltirilmaydigan maydon.

1) x^4
2) $x^4 + x + 1$
3) $x^4 + x^2 + x + 1$
4) $x^4 + x^3 + x^2 + x + 1$
5) $x^4 + 1 \rightarrow (x+1)$ ga bo'linadi.

$(a_0 + a_1 x + a_2 x^2 + x^3) \cdot (b_0 + x)$

$(a_0 + a_1 x + x^2)(b_0 + b_1 x + x^2)$

shu 2 ta ko'rinishdan biri bo'ladi.

Teorema: Faraz qilaylik $f(x) \in F[x]$, $F[x]/f(x)$ - faktor xalqa maydon bo'lishligi uchun $f(x)$-ko'phad F-maydonda keltirilmagan bo'lishligi zarur va yetarli.

Teorema. $f(x) \in F[x]$ ko'phad keltirilmaydigan 2- va 3- darajali bo'lishligi uchun $F[x]$ xalqada, $f(x)$-ko'phad F-maydonda ildizga ega bo'lmasligi zarur va yetarlidir.

Misol. Faraz qilaylik $f(x) = x^2+x+1 \in F_2[x]$. U holda faktor halqa $F_2[x]/(f)$-quyidagi elementdan $p^n = 2^2$ yoki $0, 1, x, x+1$ iborat.

"+"	0	1	x	$x+1$
0	0	1	x	$x+1$
1	1	0	$x+1$	x
x	x	$x+1$	0	1
$x+1$	$x+1$	x	1	0

"*"	0	1	x	$x+1$
0	0	0	0	0
1	0	1	x	$x+1$
x	0	x	$x+1$	1
$x+1$	0	$x+1$	1	x

$[x+1]\cdot[x] = x^2+x$

$$\begin{array}{r|l} x^2+x & x^2+x+1 \\ \cline{2-2} -\ x^2+x+1 & 1 \\ \hline -1 & \end{array}$$

$\mod 2 \quad \boxed{1}$

$[x+1]\cdot[x+1] = x^2+2x+1$

$$\begin{array}{r|l} x^2+2x+1 & x^2+x+1 \\ \cline{2-2} -\ x^2+x+1 & 1 \\ \hline \boxed{x} & \end{array}$$

Latifa

Quyidagi $F_2[x]/(x^3+x^2+x)$ faktor xalqa uchun qo'shish va ko'paytirish jadvallari tuzilsin va uning maydon tashkil etishligi tekshirilsin.

Misol 1.

$f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + x^4$, $F = F_2$

a_0	a_1	a_2	a_3
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0
1	0	0	1
1	0	1	0
1	0	1	1
1	1	0	0
1	1	0	1
1	1	1	0
1	1	1	1

$f_1(x) = x^4 = x^2 \cdot x^2 = x \cdot x^3$

$f_2(x) = x^4 + x^3 = x(x^3+x^2) = x^2(x+x^2) = x^3(x+1)$

$f_3(x) = x^4 + x^2 = x(x^3+x) = x^2(x^2+1)$

$f_4(x) = x^4 + x^3 + x^2 = x^2(x^2+x+1)$

$f_5(x) = x^4 + x = x(x^3+1)$

$f_6(x) = x^4 + x^3 + x = x \cdot (x^3+x^2+1)$

$f_7(x) = x^4 + x^2 + x = x \cdot (x^3+x+1)$

$f_8(x) = x^4 + x^3 + x^2 + x = x \cdot (x^3+x^2+x+1)$

$f_9(x) = x^4 + 1$

$f_{10}(x) = x^4 + x^3 + 1$

$f_{11}(x) = x^4 + x^2 + 1$

$f_{12}(x) = x^4 + x^3 + x^2 + 1$

$f_{13}(x) = x^4 + x + 1$

$f_{14}(x) = x^4 + x^3 + x + 1$

$f_{15}(x) = x^4 + x^2 + x + 1$

$f_{16}(x) = x^4 + x^3 + x^2 + x + 1$

$f_9(x) = x^4 + 1$

$$\begin{array}{l|l} x^4+1 & x+1 \\ \underline{x^4+x^3} & x^3+x^2+x+1 \\ -x^3+1 & \\ \text{mod}2 \Rightarrow x^3+1 & \\ \underline{x^3+x^2} & \\ -x^2+1 & \\ \text{mod}2 \Rightarrow x^2+1 & \end{array}$$

$$\begin{array}{l} x^2+1 \\ \underline{x^2+x} \\ -x+1 \\ \text{mod}2 \Rightarrow x+1 \\ \underline{x+1} \\ 0 \end{array}$$

$$f_9(x) = x^4+1 = (x+1)\cdot(x^3+x^2+x+1)$$

– $f_{10}(x) = x^4+x^3+1$

$$\begin{array}{r|l} x^4+x^3+1 & x+1 \\ -\underline{x^4+x^3} & x^3 \\ 1 & \end{array}
\qquad
\begin{array}{r|l} x^4+x^3+1 & x^2+x+1 \\ -\underline{x^4+x^3+x^2} & x^2+1 \\ x^2+1 & \\ -\underline{x^2+x+1} & \\ x & \end{array}
\qquad
\begin{array}{r|l} x^4+x^3+1 & x^2+1 \\ -\underline{x^4+x^2} & x^2+x \\ x^3+x^2+1 & \\ -\underline{x^3+x} & \\ x^2+x+1 & \\ -\underline{x^2+1} & \\ x & \end{array}$$

$f_{10}(x) = x^4+x^3+1 \Rightarrow$ keltirilmaydigan kōphad.

– $f_{11}(x) = x^4+x^2+1$

$$\begin{array}{r|l} x^4+x^2+1 & x+1 \\ -\underline{x^4+x^3} & x^3+x^2 \\ -x^3+x^2+1 & \\ \text{mod2} \Rightarrow x^3+x^2 & \\ -\underline{x^3+x^2} & \\ 1 & \end{array}
\qquad
\begin{array}{r|l} x^4+x^2+1 & x^2+x+1 \\ -\underline{x^4+x^3+x^2} & x^2+x+1 \\ x^3+1 & \\ -\underline{x^3+x^2+x} & \\ x^2+x+1 & \\ -\underline{x^2+x+1} & \\ 0 & \end{array}$$

$$f_{11}(x) = x^4+x^2+1 = (x^2+x+1)\cdot(x^2+x+1)$$

– $f_{12}(x) = x^4+x^3+x^2+1$

$$\begin{array}{r|l} x^4+x^3+x^2+1 & x+1 \\ -\underline{x^4+x^3} & x^3+x+1 \\ x^2+1 & \\ -\underline{x^2+x} & \\ -x+1 & \\ \text{mod2} \Rightarrow x+1 & \\ -\underline{x+1} & \\ 0 & \end{array}$$

$$f_{12}(x) = x^4+x^3+x^2+1 = (x+1)\cdot(x^3+x+1)$$

– $f_{13}(x) = x^4+x+1$

$$\begin{array}{r|l} x^4+x^3+x^2+x+1 & x+1 \\ \cline{2-2} -\underline{x^4+x^3} & x^3+x \\ x^2+x+1 & \\ -\underline{x^2+x} & \\ 1 & \end{array} \qquad \begin{array}{r|l} x^4+x^3+x^2+x+1 & x^2+1 \\ \cline{2-2} -\underline{x^4+x^2} & x^2+x \\ x^3+x+1 & \\ -\underline{x^3+x} & \\ 1 & \end{array}$$

$$\begin{array}{r|l} x^4+x^3+x^2+x+1 & x^2+x+1 \\ \cline{2-2} -\underline{x^4+x^3+x^2} & x^2 \\ x+1 & \end{array}$$

$f_{16}(x)=x^4+x^3+x^2+x+1 \rightarrow$ keltirilmaydigan kophad.

Misol 2.

$F_2[x]/(x^3+x^2+x)$

$p^n=2^3=8$

$[0], [1], [x], [x+1], [x^2], [x^2+1], [x^2+x], [x^2+x+1]$

"+"	0	1	x	$x+1$	x^2	x^2+1	x^2+x	x^2+x+1
0	0	1	x	$x+1$	x^2	x^2+1	x^2+x	x^2+x+1
1	1	0	$x+1$	x	x^2+1	x^2	x^2+x+1	x^2+x
x	x	$x+1$	0	1	x^2+x	x^2+x+1	x^2	x^2+1
$x+1$	$x+1$	x	1	0	x^2+x+1	x^2+x	x^2+1	x^2
x^2	x^2	x^2+1	x^2+x	x^2+x+1	0	1	x	$x+1$
x^2+1	x^2+1	x^2	x^2+x+1	x^2+x	1	0	$x+1$	x
x^2+x	x^2+x	x^2+x+1	x^2	x^2+1	x	$x+1$	0	1
x^2+x+1	x^2+x+1	x^2+x	x^2+1	x^2	$x+1$	x	1	0

"x"	0	1	x	$x+1$	x^2	x^2+1	x^2+x	x^2+x+1
0	0	0	0	0	0	0	0	0
1	0	1	x	$x+1$	x^2	x^2+1	x^2+x	x^2+x+1
x	0	x	x^2	x^2+x	x^2+x	x^2	x	0
$x+1$	0	$x+1$	x^2+x	x^2+1	x^2	1	x^2	x^2+x+1
x^2	0	x^2	x^2+x	x^2	x	x^2+x	x^2	0
x^2+1	0	x^2+1	x^2	1	x^2+x	$x+1$	x	x^2+x+1
x^2+x	0	x^2+x	x	x^2	x^2	x	x^2+x	0
x^2+x+1	0	x^2+x+1	0	x^2+x+1	0	x^2+x+1	0	x^2+x+1

$[x]\cdot[x^2] = x^3$

$$\begin{array}{r|l} x^3 & x^3+x^2+x \\ \underline{-\;x^3+x^2+x} & 1 \\ -x^2-x & \end{array}$$

mod2 $\rightarrow$ x^2+x

$x\cdot(x^2+1) = x^3+x$

$$\begin{array}{r|l} x^3+x & x^3+x^2+x \\ \underline{-\;x^3+x^2+x} & 1 \\ [x^2] & \end{array}$$

$x\cdot(x^2+x) = x^3+x^2$

$$\begin{array}{r|l} x^3+x^2 & x^3+x^2+x \\ \underline{-\;x^3+x^2+x} & 1 \\ [x] & \end{array}$$

$x(x^2+x+1) = x^3+x^2+x$

$(x+1)\cdot(x+1) = x^2+2x+1$

mod 2 $\rightarrow$ x^2+1

$(x+1)\cdot x^2 = x^3+x$

$$\begin{array}{r|l} x^3+x & x^3+x^2+x \\ \underline{-\;x^3+x^2+x} & 1 \\ [x^2] & \end{array}$$

$(x+1)(x^2+1) = x^3+x^2+x+1$

$$\begin{array}{r|l} x^3+x^2+x+1 & x^3+x^2+x \\ \underline{-\;x^3+x^2+x} & 1 \\ [1] & \end{array}$$

$(x+1)(x^2+x) = x^3+x^2+x^2+x$

mod2 $\rightarrow$ x^3+x

$(x^3+x)\ (\mathrm{mod}\ (x^3+x^2+x)) = x^2$

$(x+1)(x^2+x+1) = x^3+x^2+x^2+x+x+1$

mod 2 $\rightarrow$ x^3+1

$(x^3+1)\ (\mathrm{mod}(x^3+x^2+x)) = x^2+x+1$

$$x^2 \cdot x^2 = x^4$$

$$\begin{array}{l|l} x^4 & x^3+x^2+x \\ \hline -\,x^4+x^3+x^2 & x+1 \\ x^3+x^2 & \\ -\,x^3+x^2+x & \\ [x] & \end{array}$$

$$x^2 \cdot (x^2+1) = x^4+x^2$$

$$\begin{array}{l|l} x^4+x^2 & x^3+x^2+x \\ \hline -\,x^4+x^3+x^2 & x+1 \\ x^3 & \\ -\,x^3+x^2+x & \\ [x^2+x] & \end{array}$$

$$x^2 \cdot (x^2+x) = x^4+x^3$$

$$\begin{array}{l|l} x^4+x^3 & x^3+x^2+x \\ \hline -\,x^4+x^3+x^2 & x \\ [x^2] & \end{array}$$

$$x^2 \cdot (x^2+x+1) = x^4+x^3+x^2$$

$$x^4+x^3+x^2 = x \cdot (x^3+x^2+x)$$

$$(x^2+1)(x^2+1) = x^4+2x^2+1$$

$$\text{mod } 2 \rightarrow x^4+1$$

$$\begin{array}{l|l} x^4+1 & x^3+x^2+x \\ \hline -\,x^4+x^3+x^2 & x+1 \\ x^3+x^2+1 & \\ -\,x^3+x^2+x & \\ [x+1] & \end{array}$$

$$(x^2+1) \cdot (x^2+x) = x^4+x^3+x^2+x$$

$$\begin{array}{l|l} x^4+x^3+x^2+x & x^3+x^2+x \\ \hline -\,x^4+x^3+x^2 & x \\ [x] & \end{array}$$

$$(x^2+1)(x^2+x+1) =$$

$$= x^4+\cancel{x^2}+x^3+x+\cancel{x^2}+1 =$$

$$= x^4+x^3+x+1$$

$$\begin{array}{l|l} x^4+x^3+x+1 & x^3+x^2+x \\ \hline -\,x^4+x^3+x^2 & x \\ [x^2+x+1] & \end{array}$$

$$(x^2+x)(x^2+x) = x^4+x^3+x^3+x^2$$

$$\text{mod } 2 \rightarrow x^4+x^2$$

$$\begin{array}{l|l} x^4+x^2 & x^3+x^2+x \\ \hline -\,x^4+x^3+x^2 & x+1 \\ x^3 & \\ -\,x^3+x^2+x & \\ [x^2+x] & \end{array}$$

$$(x^2+x)(x^2+x+1) =$$

$$= x^4+\cancel{x^3}+\cancel{x^3}+\cancel{x^2}+\cancel{x^2}+x$$

$$= x^4+x$$

$$\begin{array}{l|l} x^4+x & x^3+x^2+x \\ \hline -\,x^4+x^3+x^2 & x+1 \\ x^3+x^2+x & \\ -\,x^3+x^2+x & \\ [0] & \end{array}$$

$$(x^2+x+1)(x^2+x+1) = x^4+\cancel{x^3}+x^2+\cancel{x^3}+\cancel{x^2}+\cancel{x}+$$

$$+\,x^2+\cancel{x}+1 = x^4+x^2+1$$

$$\begin{array}{l|l} x^4+x^2+1 & x^3+x^2+x \\ \hline -\,x^4+x^3+x^2 & x+1 \\ x^3+1 & \\ -\,x^3+x^2+x & \\ [x^2+x+1] & \end{array}$$

22.11.2023.

$F_q[x]$ - halqada keltirilmaydigan n-darajali kophadlar sonini aniqlash.

Myobius funksiyasi.

Ta'rif: Myobius funksiyasi deb, shunday $\mu(n)$, $n \in N$-ga

$$\mu(n)=\begin{cases} 1, & \text{agar } n=1 \\ (-1)^k, & \text{agar } n\text{-soni } k\text{-ta tub sonlar kopaytmasidan iborat bolsa} \\ 0, & \text{agar } n\text{-soni biror tub son kvadratiga bolinsa} \end{cases}$$

(3-shart bajarilmasa)

Quyidagicha belgilash kiritaylik:

$\sum_{d/n}$ - bu $n \in N$ - sonining barcha d-butun boluvchilari boyicha yigindi.

Teorema 1. $F_q[x]$ -halqadagi n-darajali normallashgan keltirilmaydigan kophadlar soni:

$$\boxed{N_q(n)=\frac{1}{n}\sum_{d/n}\mu\left(\frac{n}{d}\right)q^d=\frac{1}{n}\sum_{d/n}\mu(d)q^{n/d}}$$

Misol 1. $F_2[x]$ - xalqadagi 4-darajali normallashgan keltirilmaydigan kophadlar soni topilsin.

Yechish: $q=2$, $n=4$, $d=\{1,2,4\}$

$$N_q(4)=\frac{1}{4}\left(\mu\left(\frac{4}{1}\right)\cdot 2^1+\mu\left(\frac{4}{2}\right)\cdot 2^2+\mu\left(\frac{4}{4}\right)\cdot 2^4\right)=$$

$$=\frac{1}{4}\left(\mu(4)\cdot 2+\mu(2)\cdot 4+\mu(1)\cdot 2^4\right)=$$

$$=\frac{1}{4}\left(0\cdot 2+(-1)\cdot 4+1\cdot 2^4\right)=\frac{1}{4}\cdot(16-4)=3$$

GSM da A5 algoritmi qollanilgan.

Teorema 2. $F_q[x]$ - xalqadagi n-darajali keltirilmaydigan (normallashgan) kophadlar kopaytmasi quyidagi formula orqali topiladi.

$$\boxed{I(q;n;x)=\prod_{d|n}\left(x^{q^d}-x\right)^{\mu(n/d)}=\prod_{d|n}\left(x^{q^{n/d}}-x\right)^{\mu(d)}}$$

Vazifa: ~~F_{2^8}~~ F_{2^8} maydonda 8-darajali keltirilmaydigan kophadlar soni nechta ($n=8$ yani $N_{2^8}(8)=?$)

Yechish: (1) $n=8$ $\qquad N_{2^8}(8)=?$

$$N_{2^8}(8)=\frac{1}{8}\Big(\underset{\underset{1}{\parallel}}{\mu(1)}\cdot(2^8)^{8/1}+\underset{\underset{-1}{\parallel}}{\mu(2)}\cdot(2^8)^{8/2}+\underset{\underset{0}{\parallel}}{\mu(4)}\cdot(2^8)^{8/4}+\underset{\underset{0}{\parallel}}{\mu(8)}\cdot(2^8)^{8/8}\Big)=$$

$$=\frac{1}{8}(2^{64}-2^{32})=2^{61}-2^{29}$$

Javob: $2^{61}-2^{29}$

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(2) $$I(q;n;x)=\prod_{d/n}\left(x^{q^{n/d}}-x\right)^{\mu(d)}$$

$q=2,\quad n=4,\quad d=\{1;2;4\}$

$$I(2;4;x)=(x^{2^4}-x)^{\mu(1)}\cdot(x^{2^2}-x)^{\mu(2)}\cdot(x^{2^1}-x)^{\mu(4)}=$$

$$=(x^{16}-x)^1\cdot(x^4-x)^{-1}\cdot(x^2-x)^0=\frac{x^{16}-x}{x^4-x}=\frac{x^{15}-1}{x^3-1}$$

$q=2,\quad n=8,\quad d=\{1;2;4;8\}$

$$I(2;8;x)=(x^{2^8}-x)^{\mu(1)}\cdot(x^{2^4}-x)^{\mu(2)}\cdot(x^{2^2}-x)^{\mu(4)}\cdot(x^{2^1}-x)^{\mu(8)}=$$

$$=(x^{256}-x)^1\cdot(x^{16}-x)^{(-1)}\cdot(x^4-x)^0\cdot(x^2-x)^0=$$

$$=\frac{x^{256}-x}{x^{16}-x}=\frac{x^{255}-1}{x^{15}-1}$$

$q=2^8,\quad n=8,\quad d=\{1;2;4;8\}$

$$I(2^8;8;x)=(x^{256^8}-x)^{\mu(1)=1}\cdot(x^{256^4}-x)^{\mu(2)=-1}\cdot(x^{256^2}-x)^{\mu(4)=0}\cdot(x^{256^1}-x)^{\mu(8)=0}=$$

$$=\frac{x^{256^8}-x}{x^{256^4}-x}$$

Javob: $\dfrac{x^{256^8}-x}{x^{256^4}-x}$

29.11.2023

Berilgan sonni tublikka tekshirish usullari.

Yevklid teoremasi: Tub sonlar toplami cheksiz.

1) 1845-yil Fransuz matematigi J.L. Bertran:

$2a > 7$ bolganda

$(a; 2a-2)$ sonlar orasida kamida 1 ta tub son yotadi.

2) Natural sonlar qatorida shunday p va $p+2$ sonlar topiladiki, ularning ikkisi ham tub son boladi. Bu sonlar egizak sonlar deb ataladi.

3) $x \in N$ **da** faqat tub sondan iborat funksiyalar mavjudmi?

$f(x) = ?$ Eyler (1707-1783)

Eyler fikri boyicha $x \in [1, 15]$ oraligidagi butun sonlar uchun $f(x) = x^2 + x + 17$

har doim tub boladi.

Agar $x \in [0; 40]$ oraligidagi butun sonlar bolsa $f(x) = x^2 - x + 41$ funksiya tub boladi.

4) K. Goldbax (matem) quyidagi mulohazani ilgari surgan:

Agar $x \in N$ bolsa, barcha qiymati tub sonlardan iborat biror ta $f(x)$ funksiya mavjud emas!

5) 1851-yilda rus matematigi Chebishev:

$$a \cdot \frac{x}{\ln x} < \pi(x) < A \cdot \frac{x}{\ln x}$$

$\pi(x)$ – x dan oshmaydigan tub son.

$a > \frac{1}{2} \ln 2$, $A < 2 \cdot \ln 2$

Maxsus korinishdagi tub sonlar.

Ta'rif: Quyidagi korinishdagi sonlar Ferma sonlari deb ataladi:

$$F_k = 2^{2^k} + 1, \quad k = 0, 1, 2, \ldots$$

Teorema: $n=F_k$ soni $k>0$ bōlganda tub bōladi faqat va faqat,

$$3^{\frac{n-1}{2}} = -1 \pmod{n}$$

ōrinli bōlsa.

Ta'rif: Quyidagi kōrinishdagi sonlar **Mersen sonlari** d-di:

$$M_p = 2^p - 1, \text{ (bu yerda } p\text{-tub son}$$

M_p tub bōlsa.

06.12.2023

Mersen sonlarining tubligini aniqlash.

Teorema: Biror p-tub sonda $(p>2)$ $M_p = 2^p - 1$ Mersen soni bōlishi u-n quyidagicha ketma-ketlikda:

$$L_0 = 4, \quad L_{j+1} = (L_j^2 - 2) \pmod{M_p},$$

$$j = 0, 1, 2, \dots, p-3,$$

ushbu

$$L_{p-2} \equiv 0 \pmod{M_p}$$

shartning bajarilishi zarur va yetarli

Eng katta Mersen soni (2023-yilgacha aniqlanganlar orasida): $M_{82589933} = 2^{82589933} - 1$ (2018-yil 7-dekabrda topilgan) 24 862 048 xonali son

Misol: $p=5 \quad M_p=2^5-1=2-1=31$

$j=0,1,2.$

$L_0=4$
$L_1=(L_0^2-2)\ (\mathrm{Mod}\ 31)=14\ (\mathrm{mod}\ 31)=14$

$L_2=(L_1^2-2)\ (\mathrm{mod}\ 31)=194\ (\mathrm{mod}\ 31)=8$

$L_3=(L_2^2-2)\ (\mathrm{mod}\ 31)=62\ (\mathrm{mod}\ 31)=0. \qquad L_3=0$

Demak M_5 - tub son.

Vazifa:

$p=13 \qquad M_p=2^{13}-1$ tubmi yoki tub emasmi?

Umumiy holda sonning tubligini aniqlash usullari.

Teorema (Lukas): Agar n-natural son uchun shunday, a-butun son mavjud bo'lib:

$$a^{n-1}\equiv 1\ (\mathrm{mod}\ n) \text{ va}$$

$$a^{n-1/p}\not\equiv 1\ (\mathrm{mod}\ n)$$

$n-1$ ning bo'luvchisi bo'lgan p-tub sonlari uchun bajarilsa, u holda n-tub son bo'ladi.

Misol: $n=13.$ $n-1=12$
$p=2,3,$

$a=2.$

$a^{13-1} \equiv 2^{12} \pmod{13} = 16^3 \pmod{13} = 1.$

$2^{(13-1)/2} \equiv 2^6 \pmod{13} = 12 \cancel{\neq 1} \pmod{13}$

~~Demak n-tub son.~~

$2^{(13-1)/3} = 2^4 = 16 \pmod{13} = 3$

Demak n-tub son.

Vazifa.

$a<13$ dagi barcha a lar topilsin.

Мавзу:

ГОСТ Р 34.10-94 Rossiya Fed. ERI standarti (1994-2000) algoritm uchun tub sonlarni generatsiya qilish usuli.

ГОСТ – SSSR standarti.

ГОСТ Р – Rossiya standarti.

ECDSA – AQSh ning 2000-yildan boshlab ERI standarti (elliptik egri chiziqqa asoslan)

ГОСТ Р 34.10-2001 ⇒ RF ning ERI standarti.

Teorema: q - toq tub son, $p = q \cdot N + 1$, bu yerda N - juft son bo'lsin. Shuningdek $p < (2q+1)^2$ va quyidagi shartlar:

1) $2^{q \cdot N} \equiv 1 \pmod N$

2) $2^N \not\equiv 1 \pmod p$

bajarilsa, u holda p - tub son.

Харик. мат. и коментю. осн. крит.

Vazifa.

$p = 61, \quad q = 5, \quad N = 12$

Yechish:

1) $p = 13 \quad M_p = 2^{13} - 1$ tub sonmi?

$L_0 = 4 \qquad j = 0, 1, \dots, 10, \quad M = 2^{13} - 1 = 8191.$

$L_1 = (L_0^2 - 2) \pmod{8191} = 14.$

$L_2 = (L_1^2 - 2) \pmod{8191} = 194$

$L_3 = (L_2^2 - 2) \pmod{8191} = 4870$

$L_4 = (L_3^2 - 2) \pmod{8191} = 3953$

$L_5 = (L_4^2 - 2) \pmod{8191} = 5970$

$L_6 = (L_5^2 - 2) \pmod{8191} = 1857$

$L_7 = (L_6^2 - 2) \pmod{8191} = 36.$

$L_8 = (L_7^2 - 2) \pmod{8191} = 1294$

$L_9 = (L_8^2 - 2) \pmod{8191} = 3470$

$L_{10} = (L_9^2 - 2) \pmod{8191} = 128$

$L_{11} = (L_{10}^2 - 2) \pmod{8191} = 0$

Demak 8191 – tub son.

2) $n = 13$, $n-1 = 12$, $p = 2, 3$ $a \in [2;11] \in N$.

$a=2$: $a^{13-1} = a^{12} = 2^{12} \pmod{13} =$
$16^3 \pmod{13}$
$a=2$ $3^3 \pmod{13} = 1$.

$p=2$: $2^{\frac{12}{2}} = 2^6 = 64 \pmod{13} = 12 \neq 1$

$p=3$: $2^{\frac{12}{3}} = 2^4 = 16 \pmod{13} = 3 \neq 1$

× $a=3$: $3^{12} \pmod{13} = 27^4 \pmod{13} = 1$

$p=2$: $3^{\frac{12}{2}} = 3^6 \pmod{13} = 27^2 \pmod{13} = 1$

× $a=4$: $4^{12} \pmod{13} = 16^6 \pmod{13} = 3^6 \pmod{13} = 1$

$p=2$: $4^{\frac{12}{2}} = 4^6 \pmod{13} = 16^3 \pmod{13} = 3^3 \pmod{13} = 1$.

× $a=5$: $5^{12} \pmod{13} = 25^6 \pmod{13} = (-1)^6 \pmod{13} = 1$

$p=2$: $5^6 \pmod{13} = 25^3 = (-1)^3 = -1 = 12 \neq 1$

$p=3$: $5^4 \pmod{13} = 25^2 = (-1)^2 = 1$

$a=6$ $a=6$: $6^{12} \pmod{13} = 36^6 = (-3)^6 = 3^6 = 27^2 = 1$

$p=2$: $6^6 \pmod{13} = 36^3 = (-3)^3 = -27 = 12 \neq 1$

$p=3$: $6^4 \pmod{13} = 36^2 \pmod{13} = (-3)^2 = 9 \neq 1$

$a=7$ $a=7$: $7^{12} \pmod{13} = 49^6 = (-3)^6 = 27^2 = 1$

$p=2$: $7^6 \pmod{13} = 49^3 \pmod{13} = (-3)^3 = -27 = 12 \neq 1$

$p=3$: $7^4 \pmod{13} = 49^2 = (-3)^2 = 9 \neq 1$

$a=8$: $8^{12} \pmod{13} = 64^6 = 12^6 = (-1)^6 = 1$

$p=2$: $8^6 \pmod{13} = 64^3 = 12^3 = (-1)^3 = -1 = 12 \neq 1$

$p=3$: $8^4 \pmod{13} = 64^2 = 12^2 = (-1)^2 = 1$

$a=9$: $9^{12} \pmod{13} = 81^6 \pmod{13} = 3^6 = 27^2 = 1$

$p=2$: $9^6 \pmod{13} = 81^3 = 3^3 = 27 = 1$

$a=10$: $10^{12} \pmod{13} = 100^6 = 9^6 = 81^3 = 3^3 = 1$

$p=2$: $10^6 \pmod{13} = 100^3 = 9^3 = 81 \cdot 9 = 3 \cdot 9 = 27 = 1$ $\qquad a=11$

$a=11$: $11^{12} \pmod{13} = (-2)^{12} = 2^{12} = 16^3 = 3^3 = 27 = 1$

$p=2$: $11^6 = (-2)^6 = 2^6 = 64 = 12 \neq 1$

$p=3$: $11^4 = (-2)^4 = 2^4 = 16 = 3 \neq 1$

Javob: $a = 2, 6, 7, 11.$

3) $p=61, \quad q=5, \quad N=12$

1) $2^{q \cdot N} \equiv 1 \pmod{p}$

$2^{5 \cdot 12} \equiv 2^{60} \pmod{61} = 1$ 1-shart bajarildi.

2) $2^N \not\equiv 1 \pmod{p}$

$2^{12} \pmod{61} = 9 \neq 1$ 2-shart bajarildi

demak p $(p=61)$ soni tub son.

13.12.2023.

Primitiv ildiz

Ta'rif: Aytaylik p – tub son va $g<p$ bo'lsin. U holda g soni p modul bo'yicha primitiv ildiz (p modul bo'yicha tashkil etuvchi) deyiladi, agar har bir b soni $(1<b<p-1)$ uchun shunday a mavjud bo'lsaki, $g^a \equiv b \pmod p$ o'rinli bo'lsa.

p – tub son.

$q_1, q_2, \dots, q_n$ – $(p-1)$ sonining tub bo'luvchilari.

g soni p modul bo'yicha primitiv ildiz bo'lishi uchun $g^{\frac{p-1}{q_i}} \pmod p \neq 1$ bo'lishi lozim. Aks holda, ya'ni biror q_i da $g^{\frac{p-1}{q_i}} \pmod p = 1$ bo'lsa?

$n=13$ bo'yicha olingan natijalar

$(2, 6, 7, 11)$ primitiv ildiz ekan.

$(3, 4, 5, 8, 9, 10)$ lar primitiv ildiz emas ekan.

Berilgan sonning tubligini aniqlashning ayrim usullari

Teorema (Poklington). Aytaylik $n \geq 3$ son va $n = Q \cdot R + 1$, bu yerda

EKUB$(Q, R) = 1$ va $Q = \prod_{j=1}^{t} q_j^{\alpha_j}$ $\rightarrow$ Q ning tub ko'paytuvchilarga yoyilmasi.

Agar har bir q_j - uchun shunday

$\exists a_j$ mavjud bo'lsa va $a_j^{n-1} \equiv 1 \pmod n$ hamda

EKUB$(a_j^{\frac{n-1}{q_j}} - 1, n) = 1$ bo'lsa, u holda

n - tub son bo'ladi.

Vazifa

Poklingtonga misol

Yechish.

$n = 7 \quad Q = 6 \quad R = 1$

$n = Q \cdot R + 1 \longrightarrow 7 = 6 \cdot 1 + 1$

EKUB$(Q, R) = (6, 1) = 1$.

$Q = 2 \cdot 3 \qquad q_1 = 2, \quad q_2 = 3$

$a = 3$

1. $a^{n-1} \equiv 1 \pmod n$.

$3^6 \pmod 7 = 9^3 \pmod 7 = 2^3 \pmod 7 = 1$.

EKUB$(3^{\frac{6}{2}} - 1, 7) = (26, 7) = 1$.

2. EKUB$(3^{\frac{6}{3}} - 1, 7) = (8, 7) = 1$.

Demak $n = 7$ tub son.

22.12.2023

Berilgan sonni tublikka tekshiruvchi ehtimollik testlari (Solovey – Shtrassen va Miller-Rabin testlari)

Berilgan natural toq n soni tub ekanligini aniqlashning ehtimollik testi quyidagi tartibda amalga oshiriladi.

$\exists a \in N$, $1 < a < n$ soni tanlanadi. va uning uchun ma'lum bir shartlarning bajarilishi tekshiriladi. Agar bu shartlarning birortasi bajarilmasa & n murakkab son boladi.

Agar qaysidir a qiymatda keltirilgan barcha shartlar bajarilsa u holda n soni tub degan xulosa kelib chiqmaydi. Biroq n soni ma'lum bir ehtimollik bilan tub deyiladi.

Teorema: Aytaylik, n- toq murakkab son. U holda quyidagi shartlarni:

1) $EKUB(a, n) = 1$

2) $a^{\frac{n-1}{2}} = J\left(\frac{a}{n}\right) \bmod n$

qanoatlantiruvchi $1 < a < n$ oraliqdagi olingan butun a-lar soni $n/2$ dan oshmaydi.

Bu yerda $J\left(\frac{a}{n}\right) \rightarrow$ Yakobi simvoli.

Solovey-Shtrassen testi

1) $1 < a \le n-1$ tasodifiy k ta a sonini olamiz, $(k < n)$

2) tasodifiy olingan k ta a larga nisbatan teoremaning 1-va 2- shartlarini tekshiramiz. O'rinli bo'lsa 3-qadamga o'tamiz.

3) U holda, n-soni $\left(1 - \frac{1}{2^k}\right) = p$ ehtimollik bilan n soni tub son deb e'lon qilinadi.

Miller-Rabin ehtimollik testi

Aytaylik n-toq murakkab son va 3 ga bo'linmasin,

$n - 1 = 2^r \cdot t$ ifoda o'rinli bo'lsin, bu yerda $r \ge 1$, t-toq son. U holda

$$a^t \equiv 1 \pmod n$$

yoki

$$a^{\frac{n-1}{2^j}} \equiv -1 \pmod n,$$

ushbu $1 \le j \le r$ oraliqdan olingan biror j uchun, shartlarni qanoatlantiruvchi $0 < a \le n-1$ oraliqdan ~~aliga~~ olingan a-lar soni $n/4$ dan oshmaydi.

Natija: Teorema dan agar k ta tasodifiy a ning qiymatlari uchun n sonning murakkab emasligi aniqlansa, u holda berilgan n soni $\left(1 - \frac{1}{4^k}\right)$ ehtimollik bilan tub son deb yuritiladi.

27.12.2023

Butun sonni tub kopaytuvchilarga ajratishning zamonaviy usullari tahlili.

Ta'rif: Agar n musbat butun son uchun quyidagi kanonik $n = p_1^{\alpha_1} \cdot p_2^{\alpha_2} \cdot \ldots \cdot p_s^{\alpha_s}$ yoyilmani topish masalasi qaralayotgan bo'lsa, u holda bunday masalaga berilgan n murakkab sonni tub kopaytuvchilarga ajratish (yoki faktorizatsiya) deyiladi.

Amalda, xususan kriptologiya fanida $n = p * q$ sonni tub kopaytuvchilarga ajratish masalasi qaraladi xolos.

Bugungi kunda faktorizatsiya masalalarida asosiy ilgarilash bosqichlari sifatida chiziqli resheta, kvadratik resheta va sonli maydon resheta usullari alohida ajratib korsatiladi.

Ferma usuli

Teorema: Aytaylik, $n>1$ toq son. Bu son murakkab son boladi faqat va faqat $\exists p,q \in Z^+$ bolib, $n=p^2-q^2$ bolsa. Bu yerda $p-q>1$.

Ferma usulining mohiyati shundan iboratki, teorema natijasiga kora $\exists p,q \in Z^+$ sonlar topish kerakki,

$n=p^2-q^2$; $p^2=n+q^2$, yoki $q^2=n+p^2 \to q^2=p^2-n$ bajarilsin. Agar $p^2=n+q^2$, $q=1,2,3,\ldots$ qiymatlar uchun $n+q^2$ biror sonning tola kvadratidan iborat bolmasa, u holda $q=(n-1)/2$ qiymat uchun $n+q^2$-ni tekshirib koramiz va biror sonning kvadratidan iborat bolsa. Aks holda n-tub son boladi.

Masalan: $n=527$

q	$n+q^2$
1	$527+1=528$
2	$527+4=531$
3	$527+9=536$
4	$527+16=543$
5	$527+25=552$
6	$527+36=563$
7	$527+49=576=24^2$

Vazifa

~~$n=3$ $\quad q_{max}=(n-1)/2=(3-1)/2=1$~~

~~$q=1 \Rightarrow p^2=n+q^2=3+1=4 \Rightarrow p=2$ $\quad n=(p-q)(p+q)=1\cdot 3$~~

$n=13$ $\qquad q_{max}=(n-1)/2=(13-1)/2=6$

q	$n+q^2$
1	$13+1=14$
2	$13+4=17$
3	$13+9=22$
4	$13+16=29$
5	$13+25=38$
6	$13+36=49$

$q=6$, $p=7$

$n=(7-6)(7+6)=1\cdot 13$

lekin $q-p>1$ shartni qanoatlantirmaydi $(7-6=1\not>1)$

11.01.2024

Ferma usulining modifikatsiyalangan varianti.

1) $n\in N$ son.

2) $q^2=p^2-n$ (1), bu yerda $p=m, m+1, \ldots$ m bu $m^2\geq n$ shartni qanoatlantiruvchi eng kichik natural son.

3) (p^2-n) to'la kvadrat bo'lguncha yoki $p=\frac{n+1}{2}$ gacha (1) ifodani hisoblaymiz.

Agar $p=m, m+1, \ldots, \frac{n+1}{2}$ oraliqda (1) ifoda to'la kvadrat bo'lmasa, demak n-tub son.

Misol.

$n=527$

$q^2=p^2-n$

$m=23 \to 529-527=2$

$m+1=24 \to 576-527=49=7^2 \to p=24$, $q=7$

$n=(24-7)(24+7)=17\cdot 31$

Mavzu: Pollardning ρ usuli

Algoritm 3 ta qadamdan iborat.

1) $f(x)$ - kophad darajasi 2 yoki undan katta bo'lgan, masalan $f(x)=x^2+1$.

2) Tasodifiy $x_0 \in Z n Z$ $(0<x_0 \le n-1)$

3) Qandaydir fiksirlangan (ma'lum bo'lgan) j, k-nomerlar uchun quyidagi shartlarning bajarilishi tekshiriladi:

$1<EKUB(x_j-x_k; n)<n$ $\qquad x_i \equiv f(x_{i-1}) \pmod n$

toki, n-sonining tub ko'paytuvchilari topilmaguncha.

Bu yerda, agar j soni $2^h \le j < 2^{h+1}$, $h \in N$ bo'lsa u holda $k=2^h-1$ ko'rinishda olish maqsadga muvofiq.

Vazifa

$n=1001$ 1) Fermaning modif. usuli
2) Pollardning ρ usuli.

Yechish.

1. Fermaning modifikatsiyalangan usuli

$n = 1001 \qquad m = [\sqrt{n}] + 1 = 31 + 1 = 32$

$p_{max} = \frac{n+1}{2} = \frac{1002}{2} = 501.$

p	$p^2 - n$	q
32	23	
33	88	
34	155	12,44
35	224	14,96
36	295	17,17
...	...	...
45	1024	32

$\rightarrow \quad p = 45 \quad q = 32$

$1001 = (45 - 32)(45 + 32)$

$1001 = 13 \cdot 77.$

keyin 13 va 77 uchun ham shu algoritm qollaniladi.

Natija: $1001 = 7 \cdot 11 \cdot 13$ (tub emas)

2. Pollardning ρ usuli.

$n = 1001 \qquad x_0 = 2 \qquad f(x) = x^2 + 1$

$x_1 = 5 \qquad x_4 = 458330 \ (\cancel{mod}\ 1001) = 873$

$x_2 = 26$

$x_3 = 677$

$(x_1 - x_0;\ n) = (5-2,\ 1001) = (3,\ 1001) = 1.$

$(x_2 - x_0,\ n) = (26-2,\ 1001) = (24,\ 1001) = 1$

$(x_2 - x_1,\ n) = (26-5,\ 1001) = (21,\ 1001) = 7$

$1001 = 7 \cdot 143$

7 va 143 uchun algoritm yana takrorlanadi.

7 - tub son.

$n = 143. \qquad x_0 = 2 \qquad f(x) = x^2 + 1$

$(x_1 - x_0,\ n) = (3,\ 143) = 1$

$(x_2 - x_0,\ n) = (24,\ 143) = 1$

$(x_2 - x_1,\ n) = (21,\ 143) = 1$

$(x_3 - x_0,\ n) = (675,\ 143) = 1$

$(x_3 - x_1, n) = (672, 143) = 1$

$(x_3 - x_2, n) = (651, 143) = 1$

$(x_4 - x_0, n) = (458328, 143) = 13.$

$x_4 = 458330 \pmod{143} =$

demak $143 = 11 \cdot 13$

11 va 13 uchun ρ usuli qaytadan qo'llaniladi va ularning tub son ekanligi kelib chiqadi.

demak: Javob: $1001 = 7 \cdot 11 \cdot 13$ (tub emas).

17.01.2024.

chekli maydonda diskret logarifmlash usullari.

Ta'rif: Agar berilgan a, b butun va $1 < a, b < p$ sonlar uchun quyidagi taqqoslama $a^x \equiv b \pmod p$ tenglama x butun yechimini topish masalasi qaralayotgan bo'lsa, u holda bunday masala F_p chekli maydonda diskret logarifmlash deb ataladi.

Bugungi kunda kriptografiya sohasidagi mutaxassislar uchun quyidagi 3 ta gruppada diskret logarifmlash masalalari dolzarb hisoblanadi.

- $GF(p)$ - tub sonlar maydoni multiplikativ gruppasi.
- $GF(2^n)$ - 2 xarakteristikali chekli maydon multiplikativ gruppasi

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Ochiq adabiyotlarda tub sonlar maydonida diskret logarifmlashning 3 ta usuli mavjud:

- Chiziqli resheta usuli.
- Gauss butun sonlar sxemasi usuli,
- Sonli maydon resheta usuli.

24.01.2024.

Chekli maydonda diskret logarifmlashning determinirlangan usuli.

Bu usul quyidagi qadamlardan iborat:

1) Quyidagi son hisoblansin:

$$H := [p^{1/2}] + 1 \qquad (a^x \equiv b \pmod p)$$

2) Quyidagi son hisoblansin:

$$C \equiv a^H \pmod p$$

3) u, $1 \le u \le H$ sonli qiymatlari uchun $C^u \pmod p$ jadval tuzing. Ushbu qiymatlarni tartiblab chiqing.

4) Keyingi jadval esa $b \cdot a^v \pmod p$, $0 \le v \le H$ qiymatlar uchun tuzilib tartiblansin.

5) Birinchi va ikkinchi jadvalda teng chiqqan u, v elementlar olinsin (Aks holda javob yo'q)

6) Javob sifatida:

$$x \equiv H \cdot u - v \pmod{p-1}$$

olinsin.

Misol.

Quyidagi $3^x \equiv 15 \pmod{17}$ ifodadan x ni toping.

Yechish: $x = 6 \rightarrow 3^6 = 729$; $729 = 42 \cdot 17 + 15$

Shuni ta'kidlash kerakki, bizni

faqat butun yechimlar qoraqtiradi:

1) $H = [p^{1/2}] + 1$ $\qquad (a=3,\ b=15,\ p=17)$

$H = [17^{1/2}] + 1 = 4+1 = 5$

2) $c \equiv a^{H} \pmod{p} = 3^5 \pmod{17} = 5$

3) $1 \le u \le H \to 1 \le u \le 5.$

$u=1 \to c^u \pmod{p} = 5^1 \pmod{17} = 5$
$u=2 \to c^u \pmod{p} = 5^2 \pmod{17} = 8$
$u=3 \to 5^3 \pmod{17} = 6$
$u=4 \to 5^4 \pmod{17} = 13$
$u=5 \to 5^5 \pmod{17} = 14$ $\qquad \{5, 6, 8, 13, 14\}$

4) $0 \le v \le H \qquad 0 \le v \le 5.$

$v=0 \to b \cdot a^v \pmod{p} = 15 \cdot 3^0 \pmod{17} = 15$
$v=1 \to 15 \cdot 3 \pmod{17} = 11.$
$v=2 \to 15 \cdot 9 \pmod{17} = 16$
$v=3 \to 15 \cdot 3^3 \pmod{17} = 14$
$v=4 \to 15 \cdot 3^4 \pmod{17} = 8$
$v=5 \to 15 \cdot 3^5 \pmod{17} = 7.$ $\qquad \{7, 8, 11, 14, 15, 16\}$

5) $u = v \to 8, 14$

6) $x \equiv H \cdot u - v \pmod{p-1}$

$x \equiv 5 \cdot 2 - 4 \pmod{16} = 6$

$x \equiv 5 \cdot 5 - 3 \pmod{16} = 6$